

Theory of Algebraic Topology 4

July 29, 2026

Lecture notes with worked examples in algebraic topology

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1 Problem 7: Free subgroups of F_2

Problem statement. Show that the free group F_2 contains subgroups isomorphic to the free group F_n for any $n > 1$.

Solution. Write

$$F_2 = \langle x, y \rangle.$$

Fix an integer $n > 1$. Define a homomorphism

$$\psi : F_2 \longrightarrow \mathbb{Z}/(n-1)\mathbb{Z}$$

by

$$\psi(x) = 1, \quad \psi(y) = 0.$$

This is surjective, so its kernel

$$H = \ker(\psi)$$

is a subgroup of index $n-1$ in F_2 .

By the Nielsen–Schreier theorem, every subgroup of a free group is itself free. Moreover, if a subgroup H has index d in a free group of rank r , then

$$\text{rank}(H) = 1 + d(r-1).$$

Here $r = 2$ and $d = n-1$, so

$$\text{rank}(H) = 1 + (n-1)(2-1) = 1 + (n-1) = n.$$

Hence H is a free group of rank n . Therefore

$$H \cong F_n.$$

So F_2 contains a subgroup isomorphic to F_n for every $n > 1$.

2 Problem 8: The wedge $\mathbb{RP}^2 \vee \mathbb{RP}^2$

Problem statement. Let $Y = \mathbb{RP}^2 \vee \mathbb{RP}^2$ and let $*$ be the wedge point.

(i) Show that

$$\pi_1(Y, *) \cong \mathbb{Z}/2 * \mathbb{Z}/2 \cong \langle a, b \mid a^2, b^2 \rangle.$$

(ii) Describe the covering space of Y corresponding to the kernel of the homomorphism

$$\phi : \langle a, b \mid a^2, b^2 \rangle \rightarrow \mathbb{Z}/2$$

given by $\phi(a) = 1$ and $\phi(b) = 0$. Hence show that $\ker(\phi)$ is isomorphic to $\langle a, b \mid a^2, b^2 \rangle$.

(iii) Draw a picture of the universal cover $\tilde{Y}$. Deduce that ab has infinite order in $\langle a, b \mid a^2, b^2 \rangle$.

Solution to (i). Since $Y = \mathbb{RP}^2 \vee \mathbb{RP}^2$ is a wedge of two path-connected spaces, van Kampen's theorem gives

$$\pi_1(Y, *) \cong \pi_1(\mathbb{RP}^2) * \pi_1(\mathbb{RP}^2).$$

Now

$$\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2,$$

so

$$\pi_1(Y, *) \cong \mathbb{Z}/2 * \mathbb{Z}/2.$$

If a and b denote the nontrivial loops in the two copies of $\mathbb{RP}^2$, then this free product has presentation

$$\pi_1(Y, *) \cong \langle a, b \mid a^2 = 1, b^2 = 1 \rangle.$$

Equivalently,

$$\pi_1(Y, *) \cong \langle a, b \mid a^2, b^2 \rangle.$$

Solution to (ii). Let

$$G = \langle a, b \mid a^2, b^2 \rangle \cong \mathbb{Z}/2 * \mathbb{Z}/2,$$

and consider the surjective homomorphism

$$\phi : G \rightarrow \mathbb{Z}/2, \quad \phi(a) = 1, \quad \phi(b) = 0.$$

Its kernel $H = \ker(\phi)$ has index 2, so the corresponding covering space is a connected 2-sheeted cover of Y .

We now describe this cover geometrically. Over the first copy of $\mathbb{RP}^2$, corresponding to the generator a , the map $\phi(a) = 1$ means that this summand is lifted by its nontrivial double cover

$$S^2 \longrightarrow \mathbb{RP}^2.$$

Let $p, q \in S^2$ be the two points lying over the wedge point $*$ in $\mathbb{RP}^2$.

Over the second copy of $\mathbb{RP}^2$, corresponding to the generator b , we have $\phi(b) = 0$. This means that this summand does not unwrap; instead, there are two separate copies of $\mathbb{RP}^2$, one attached at p and one attached at q .

Thus the covering space X corresponding to $\ker(\phi)$ is obtained by taking:

- one sphere S^2 ,
- one copy of $\mathbb{RP}^2$ attached to S^2 at p ,
- another copy of $\mathbb{RP}^2$ attached to S^2 at q .

The covering map $X \rightarrow Y$ is:

- the standard double cover $S^2 \rightarrow \mathbb{RP}^2$ on the first summand,
- a homeomorphism from each attached copy of $\mathbb{RP}^2$ onto the second summand.

Now compute the fundamental group of X . Since S^2 is simply connected, it contributes nothing to the fundamental group. So

$$\pi_1(X) \cong \pi_1(\mathbb{RP}^2) * \pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2 * \mathbb{Z}/2 \cong \langle u, v \mid u^2, v^2 \rangle.$$

But $\pi_1(X)$ is precisely the subgroup corresponding to the cover, namely $\ker(\phi)$. Hence

$$\ker(\phi) \cong \mathbb{Z}/2 * \mathbb{Z}/2 \cong \langle a, b \mid a^2, b^2 \rangle.$$

So $\ker(\phi)$ is isomorphic to the same abstract group again.

Solution to (iii). The universal cover $\tilde{Y}$ may be understood from the fact that

$$\pi_1(Y) \cong \mathbb{Z}/2 * \mathbb{Z}/2.$$

The Bass–Serre tree of this free product is an infinite line. Over each vertex sits a copy of the universal cover of $\mathbb{RP}^2$, namely S^2 . Therefore $\tilde{Y}$ looks like an infinite chain of 2-spheres, each touching the next one in a single point:

$$\dots - S^2 - S^2 - S^2 - S^2 - \dots$$

where adjacent spheres meet at one point, and the spheres alternate according to the two $\mathbb{RP}^2$ -summands.

The generators a and b each act as involutions, while the product ab acts as a translation along this infinite chain. A nonzero power $(ab)^m$ therefore moves a chosen lift of the basepoint to a different point of the universal cover. Hence

$$(ab)^m \neq 1 \quad \text{for all } m > 0.$$

So ab has infinite order in

$$\langle a, b \mid a^2, b^2 \rangle.$$

3 Problem 9: A cell structure for the Klein bottle

Problem statement. Show that the Klein bottle has a cell structure with a single 0-cell, two 1-cells, and a single 2-cell. Deduce that its fundamental group has a presentation

$$\langle a, b \mid baba^{-1} \rangle,$$

and show this is isomorphic to the group G in Q13 of Example Sheet 1.

Solution. Take the usual square model of the Klein bottle. In this model the boundary edges are identified in pairs, with one pair identified in the same direction and the other pair identified in opposite directions.

Because all four vertices of the square are identified together, there is exactly one 0-cell.

The two edge-identification classes give two 1-cells, which we denote by a and b .

The interior of the square gives one 2-cell.

Therefore the Klein bottle has a CW-structure consisting of:

$$1 \text{ zero-cell}, \quad 2 \text{ one-cells}, \quad 1 \text{ two-cell}.$$

Now compute the fundamental group from this cell structure. The 1-skeleton is a wedge of two circles, so its fundamental group is the free group on two generators:

$$\pi_1(K^{(1)}) \cong \langle a, b \rangle.$$

The attaching map of the 2-cell is read off from the boundary of the square. This gives the relation

$$baba^{-1} = 1,$$

up to cyclic permutation of the boundary word. Hence van Kampen gives

$$\pi_1(K) \cong \langle a, b \mid baba^{-1} \rangle.$$

This presentation is equivalent to the standard one for the Klein bottle group. Indeed,

$$baba^{-1} = 1$$

implies

$$bab = a,$$

and rearranging gives the familiar relation

$$aba^{-1} = b^{-1}.$$

Equivalently,

$$ab = b^{-1}a.$$

So we may also write

$$\pi_1(K) \cong \langle a, b \mid aba^{-1} = b^{-1} \rangle.$$

This shows that the Klein bottle group is the semidirect product

$$\mathbb{Z} \rtimes \mathbb{Z},$$

where one copy of $\mathbb{Z}$ acts on the other by inversion. Explicitly,

$$\pi_1(K) \cong \mathbb{Z} \rtimes_{(-1)} \mathbb{Z}.$$

If the group G in Q13 of Example Sheet 1 is given in this semidirect-product form, then the isomorphism is obtained by sending

$$a \mapsto (0, 1), \quad b \mapsto (1, 0).$$

Under this identification,

$$aba^{-1} = b^{-1},$$

which is exactly the Klein bottle relation. Hence

$$\pi_1(K) \cong G.$$

4 Theory: free groups, coverings, wedge sums, and the Klein bottle

These three problems are tied together by one central principle of algebraic topology:

$$\text{geometry of a space} \longleftrightarrow \text{algebra of its fundamental group}.$$

Problem 7 is about free groups and their subgroups. Problem 8 is about wedge sums, free products, and covering spaces. Problem 9 is about CW-structures and how 2-cells produce relations in π_1 .

The purpose of this section is to explain the general theory behind these examples.

4.1 Free groups and why they are fundamental

A *free group* on a set S is the group whose elements are reduced words in the letters of S and their formal inverses, with multiplication given by concatenation followed by cancellation. The only relations are the trivial cancellations

$$xx^{-1} = x^{-1}x = 1.$$

If $|S| = n$, we denote the free group by F_n .

Thus:

$$F_1 \cong \mathbb{Z}, \quad F_2 = \langle x, y \rangle,$$

and more generally F_n is the free group on n generators.

The word “free” means exactly that there are no extra relations among the generators. For example, in $F_2 = \langle x, y \rangle$, the words

$$x, \quad y, \quad xy, \quad xyx^{-1}y, \quad x^2y^{-3}xy$$

are all distinct unless one can reduce one to another by obvious cancellations.

Free groups are fundamental because they appear naturally as fundamental groups of graphs. If a connected graph has one vertex and n loop-edges, then it is a wedge of n circles and its fundamental group is F_n . In particular,

$$\pi_1\left(\bigvee_{i=1}^n S^1\right) \cong F_n.$$

So free groups arise as the basic “unconstrained” fundamental groups of 1-dimensional spaces.

4.2 Subgroups of free groups: the Nielsen–Schreier theorem

One of the deepest and most beautiful facts about free groups is the following.

Theorem (Nielsen–Schreier). Every subgroup of a free group is itself free.

This is remarkable because, in general group theory, subgroups of groups with a given structure need not preserve that structure in such a clean way. But for free groups, every subgroup is again free, although its rank may change.

There is also a precise formula when the subgroup has finite index.

Theorem (rank formula). Let $H \leq F_r$ be a subgroup of finite index d . Then

$$\text{rank}(H) = 1 + d(r - 1).$$

In particular, if $H \leq F_2$ has index d , then

$$\text{rank}(H) = 1 + d.$$

So if one constructs a subgroup of index $n - 1$ inside F_2 , one automatically gets a free group of rank

$$1 + (n - 1) = n.$$

This is exactly the mechanism used in Problem 7.

Geometric meaning. Since

$$F_r \cong \pi_1\left(\bigvee_{i=1}^r S^1\right),$$

a subgroup $H \leq F_r$ corresponds to a connected covering graph of the wedge of r circles. But every connected graph again has free fundamental group. Thus the Nielsen–Schreier theorem has a natural topological explanation: subgroups correspond to coverings, and coverings of graphs are graphs.

4.3 Wedge sums and free products

If (X, x_0) and (Y, y_0) are based spaces, their *wedge sum* is the space

$$X \vee Y = (X \sqcup Y)/(x_0 \sim y_0),$$

obtained by identifying the chosen basepoints.

This construction is especially important for fundamental groups because of the Seifert–van Kampen theorem.

Theorem (van Kampen, wedge form). If X and Y are path-connected based spaces, then

$$\pi_1(X \vee Y) \cong \pi_1(X) * \pi_1(Y),$$

where $*$ denotes the free product of groups.

The free product $G * H$ is the group formed by alternating words in elements of G and H , with only the relations already present inside G and inside H . It is the group-theoretic analogue of putting two spaces together at one point.

A basic example is

$$\mathbb{Z}/2 * \mathbb{Z}/2 \cong \langle a, b \mid a^2 = 1, b^2 = 1 \rangle.$$

This group is infinite, even though both a and b individually have order 2. The reason is that there is no relation forcing the mixed word ab to have finite order.

This is the group appearing in Problem 8, because

$$\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2,$$

and therefore

$$\pi_1(\mathbb{RP}^2 \vee \mathbb{RP}^2) \cong \mathbb{Z}/2 * \mathbb{Z}/2.$$

4.4 The group $\mathbb{Z}/2 * \mathbb{Z}/2$ and the infinite dihedral group

The free product

$$\mathbb{Z}/2 * \mathbb{Z}/2 = \langle a, b \mid a^2 = b^2 = 1 \rangle$$

is one of the simplest nontrivial examples in combinatorial group theory. It is isomorphic to the infinite dihedral group D_∞ .

A useful way to think about it is to imagine a and b as reflections of the real line. Each reflection has order 2, but the product of two reflections is a translation. A translation has infinite order unless it is trivial.

Thus in

$$\langle a, b \mid a^2, b^2 \rangle$$

the element ab has infinite order.

This is exactly what appears in Problem 8(iii). Although $a^2 = b^2 = 1$, there is no contradiction in ab having infinite order. In fact this is the generic behavior in a free product: relations inside the factors do not create unexpected relations between mixed words.

4.5 Covering spaces and subgroups of the fundamental group

One of the main bridges between topology and group theory is the classification of covering spaces.

Theorem (classification of covering spaces). Let X be connected, locally path-connected, and semilocally simply connected. Then connected covering spaces of X , up to isomorphism, correspond to subgroups of $\pi_1(X, x_0)$, up to conjugacy.

If

$$p : \widetilde{X} \rightarrow X$$

is a connected covering and $\tilde{x}_0 \in p^{-1}(x_0)$, then the induced subgroup

$$p_*\pi_1(\widetilde{X}, \tilde{x}_0) \leq \pi_1(X, x_0)$$

is the subgroup corresponding to the cover.

Conversely, given a subgroup $H \leq \pi_1(X, x_0)$, there exists a connected covering space $X_H \rightarrow X$ such that

$$p_*\pi_1(X_H) \cong H.$$

If H is normal, then the covering is regular, and the deck transformation group is

$$\text{Deck}(X_H/X) \cong \pi_1(X, x_0)/H.$$

This is precisely the theory used in Problem 8: the kernel of a homomorphism

$$\phi : \pi_1(Y) \rightarrow \mathbb{Z}/2$$

is a normal subgroup of index 2, so it determines a connected 2-sheeted regular covering of Y .

4.6 How homomorphisms to finite groups produce finite-sheeted covers

Suppose X is a nice connected space and

$$\phi : \pi_1(X, x_0) \rightarrow G$$

is a surjective homomorphism onto a finite group G . Then the kernel

$$H = \ker(\phi)$$

has finite index $|G|$, and therefore corresponds to a connected $|G|$ -sheeted covering of X .

This is often the easiest way to construct finite-sheeted covers: rather than directly building the space, one first writes down a quotient of the fundamental group.

Geometrically, loops behave according to how they map under ϕ :

- if a loop represents an element of $\ker(\phi)$, then it lifts to a closed loop in the cover;
- if it maps nontrivially, then lifting it can move a point from one sheet to another.

In Problem 8, this explains the form of the 2-sheeted cover of

$$Y = \mathbb{RP}^2 \vee \mathbb{RP}^2.$$

The generator a maps nontrivially to $\mathbb{Z}/2$, so the first $\mathbb{RP}^2$ is replaced by its double cover S^2 . The generator b maps trivially, so the second $\mathbb{RP}^2$ appears as two separate copies, one over each sheet.

This gives a very concrete example of how algebraic data in π_1 controls the topology of the corresponding covering space.

4.7 Universal covers and actions of π_1

A *universal cover* of a connected space X is a simply connected covering space

$$p : \widetilde{X} \rightarrow X.$$

When X is connected, locally path-connected, and semilocally simply connected, such a cover exists and is unique up to isomorphism.

The universal cover is extremely important because the fundamental group acts on it by deck transformations. In fact,

$$\text{Deck}(\widetilde{X}/X) \cong \pi_1(X, x_0).$$

Thus one can often study $\pi_1(X)$ by studying how it acts on $\widetilde{X}$. This is the geometric meaning of many group-theoretic phenomena.

In Problem 8, the universal cover of

$$\mathbb{RP}^2 \vee \mathbb{RP}^2$$

is built from copies of the universal cover of $\mathbb{RP}^2$, namely S^2 , arranged according to the combinatorics of the free product

$$\mathbb{Z}/2 * \mathbb{Z}/2.$$

The result is an infinite chain of spheres, each meeting the next in a point.

The element ab acts by shifting this chain along itself. Since such a shift can never have finite order, this immediately shows that ab has infinite order. This illustrates a general principle:

Principle. To understand an element of $\pi_1(X)$, study its action on the universal cover $\widetilde{X}$.

If an element acts trivially, it is the identity. If it translates points along an infinite pattern, then it must have infinite order.

4.8 CW-complexes and presentations of π_1

A CW-complex is built by attaching cells of increasing dimension:

- 0-cells are points,
- 1-cells are intervals attached along endpoints,
- 2-cells are disks attached along boundary circles,
- and so on.

For the fundamental group, the most important part is the 2-skeleton. The general rule is:

- the 1-cells give generators;
- the 2-cells give relations.

Indeed, if a connected CW-complex has one 0-cell and 1-cells

$$a_1, \dots, a_n,$$

then the 1-skeleton is a wedge of n circles, so

$$\pi_1(X^{(1)}) \cong F_n.$$

Each 2-cell is attached by a loop in the 1-skeleton, and that attaching loop yields a relation in the fundamental group. Therefore, if the attaching words are $r_1, \dots, r_m$, then

$$\pi_1(X) \cong \langle a_1, \dots, a_n \mid r_1, \dots, r_m \rangle.$$

This is the general mechanism behind polygonal presentations of surface groups. The cell structure converts topology directly into a group presentation.

4.9 Polygonal models of surfaces

Many surfaces can be obtained by identifying edges of a polygon. This point of view is particularly effective for computing fundamental groups.

For example:

- the torus is obtained from a square with opposite sides identified in the same direction;
- the Klein bottle is obtained from a square with one pair identified in the same direction and the other pair in opposite directions;
- the projective plane is obtained from a disk by identifying antipodal boundary points.

In each case:

1. the identified vertices give the 0-cells,
2. the edge classes give the 1-cells,
3. the interior of the polygon gives the 2-cell,
4. the boundary word of the polygon gives the relation in π_1 .

Thus quotient pictures of surfaces naturally lead to cell structures, and cell structures naturally lead to presentations of the fundamental group.

4.10 The Klein bottle and its fundamental group

The Klein bottle K is a standard example of a nonorientable compact surface. It can be represented as a square whose boundary identifications produce:

- one 0-cell, because all vertices are identified;
- two 1-cells, corresponding to the two edge classes;
- one 2-cell, coming from the interior of the square.

Hence the 1-skeleton is a wedge of two circles, and

$$\pi_1(K^{(1)}) \cong F_2 = \langle a, b \rangle.$$

The boundary of the square gives the attaching word of the 2-cell, so one obtains a presentation of the form

$$\pi_1(K) \cong \langle a, b \mid \text{one relation} \rangle.$$

In the standard convention, the Klein bottle group is

$$\pi_1(K) \cong \langle a, b \mid aba^{-1} = b^{-1} \rangle.$$

Equivalently,

$$ab = b^{-1}a.$$

This relation means that conjugation by a reverses b . So the group is not abelian, but it is still very structured.

Indeed, it can be written as a semidirect product

$$\pi_1(K) \cong \mathbb{Z} \rtimes_{(-1)} \mathbb{Z},$$

where one copy of $\mathbb{Z}$ acts on the other by inversion.

This semidirect-product description is extremely useful. It shows that the Klein bottle group is built from two infinite cyclic groups, but one twists the other by the automorphism

$$n \mapsto -n.$$

4.11 Normal forms in the Klein bottle group

Because of the relation

$$ab = b^{-1}a,$$

one can move powers of a and b past one another in a controlled way. As a result, every element of the Klein bottle group can be written in a normal form such as

$$a^m b^n \quad \text{or} \quad b^n a^m,$$

depending on the chosen convention.

This makes the group much more manageable than an arbitrary presented group. It is not free, because it has a nontrivial relation. It is not abelian, because a does not commute with b . But it is still algebraically accessible because the relation tells us exactly how conjugation works.

Thus the Klein bottle group is one of the first important examples of a group that is:

- nonabelian,
- defined by a simple geometric quotient,
- naturally expressible as a semidirect product.

4.12 Universal cover of the Klein bottle

Although the Klein bottle is nonorientable, its universal cover is still the plane:

$$\widetilde{K} \cong \mathbb{R}^2.$$

This is similar to the torus, whose universal cover is also $\mathbb{R}^2$. But the deck transformation groups are different.

For the torus, the deck transformation group is $\mathbb{Z}^2$, acting by translations. For the Klein bottle, the deck transformations include a twist corresponding to the relation

$$aba^{-1} = b^{-1}.$$

So two spaces can have the same universal cover but different fundamental groups, depending on how the deck transformations act. This is a very important lesson in covering space theory.

4.13 How Problems 7–9 fit together

These problems form a coherent progression.

Problem 7. Free groups arise as fundamental groups of graphs. Their subgroups correspond to covering graphs. Hence subgroups of free groups are again free.

Problem 8. Wedge sums produce free products of groups via van Kampen. Subgroups of these groups correspond to covering spaces, and the universal cover reveals algebraic properties such as infinite order of certain elements.

Problem 9. A cell structure gives generators and relations for the fundamental group. In the case of the Klein bottle, a quotient square yields a one-relator presentation and a semidirect-product description.

So the common thread is:

space $\longrightarrow$ decomposition or cover $\longrightarrow$ fundamental group $\longrightarrow$ presentation or subgroup structure.

This is one of the main organizing ideas of algebraic topology and geometric group theory.

4.14 Summary of the key ideas

For convenience, we collect the main facts used in Problems 7–9.

- A wedge of n circles has fundamental group F_n .
- Every subgroup of a free group is free.
- If $H \leq F_r$ has finite index d , then

$$\text{rank}(H) = 1 + d(r - 1).$$

- For based path-connected spaces,

$$\pi_1(X \vee Y) \cong \pi_1(X) * \pi_1(Y).$$

- Since $\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2$,

$$\pi_1(\mathbb{RP}^2 \vee \mathbb{RP}^2) \cong \mathbb{Z}/2 * \mathbb{Z}/2.$$

- Connected covers correspond to subgroups of π_1 .
- The universal cover is simply connected, and π_1 acts on it by deck transformations.
- In a CW-complex, 1-cells give generators and 2-cells give relations in π_1 .
- The Klein bottle group has presentation

$$\langle a, b \mid aba^{-1} = b^{-1} \rangle,$$

and is isomorphic to

$$\mathbb{Z} \rtimes_{(-1)} \mathbb{Z}.$$

These facts are the theoretical backbone of Problems 7–9.

4.15 Graphs as topological spaces

A *graph* in algebraic topology is usually viewed as a 1-dimensional CW-complex. This means:

- the 0-cells are called *vertices*,
- the 1-cells are called *edges*,
- each edge is attached by sending its two endpoints to vertices.

So a graph is not just a combinatorial object, but also a topological space obtained by gluing intervals together.

The most important examples are:

- the circle S^1 , which is a graph with one vertex and one loop-edge;
- the wedge $S^1 \vee S^1$, which is a graph with one vertex and two loop-edges;
- more generally, the wedge of n circles

$$\bigvee_{i=1}^n S^1,$$

often called the *rose with n petals*.

These are fundamental because

$$\pi_1(S^1) \cong \mathbb{Z}, \quad \pi_1(S^1 \vee S^1) \cong F_2, \quad \pi_1\left(\bigvee_{i=1}^n S^1\right) \cong F_n.$$

Thus graphs are the geometric source of free groups.

4.16 Rank of the fundamental group of a connected graph

Let Γ be a finite connected graph with V vertices and E edges. Then

$$\pi_1(\Gamma)$$

is a free group, and its rank is

$$\text{rank } \pi_1(\Gamma) = E - V + 1.$$

This number is often called the *first Betti number* of the graph.

Reason. Choose a spanning tree $T \subseteq \Gamma$. Since a tree is contractible, collapsing T to a point does not change the homotopy type except that every edge not in the tree becomes a loop. The number of such remaining edges is exactly

$$E - (V - 1) = E - V + 1.$$

Hence Γ is homotopy equivalent to a wedge of $E - V + 1$ circles, so

$$\pi_1(\Gamma) \cong F_{E-V+1}.$$

This gives a very concrete way to compute free ranks from pictures.

4.17 Trees and why they matter

A *tree* is a connected graph with no cycles. Topologically, every tree is contractible. Therefore

$$\pi_1(T) = 1$$

for every tree T .

Trees are important because they form the simply connected 1-dimensional spaces, so they play the role of universal covers of graphs. For example:

- the universal cover of S^1 is the infinite line;
- the universal cover of $S^1 \vee S^1$ is an infinite tree.

This is the geometric reason that free groups act naturally on trees.

4.18 Covering maps of graphs

A map

$$p : \tilde{\Gamma} \rightarrow \Gamma$$

between graphs is a *covering map* if each point of Γ has a neighborhood evenly covered by p . For graphs, this means locally the map looks like a disjoint union of copies of the same small star-shaped neighborhood.

At the level of vertices, a covering of graphs has a very simple local form: around each vertex $\tilde{v} \in \tilde{\Gamma}$, the incident edges map bijectively onto the incident edges at $p(\tilde{v}) \in \Gamma$, respecting how the edges are attached.

Thus a covering graph is obtained by repeating the local pattern of the base graph, possibly infinitely many times.

4.19 The basic example: the universal cover of the circle

The universal cover of the circle is

$$p : \mathbb{R} \rightarrow S^1, \quad p(t) = e^{2\pi it}.$$

If we draw S^1 as a loop based at one vertex, then its universal cover is the infinite line:

$$\cdots \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \cdots$$

Every edge in the line maps homeomorphically onto the loop-edge of the circle.

Since $\mathbb{R}$ is contractible,

$$\pi_1(S^1) \cong \text{Deck}(\mathbb{R}/S^1) \cong \mathbb{Z}.$$

The generator acts by translation

$$t \mapsto t + 1.$$

So the infinite line is the covering graph encoding the group $\mathbb{Z}$.

4.20 The universal cover of $S^1 \vee S^1$

Let

$$R_2 = S^1 \vee S^1$$

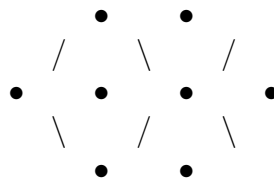
be the rose with two petals, with loop-edges labeled a and b . Then

$$\pi_1(R_2) \cong F_2 = \langle a, b \rangle.$$

Its universal cover is an infinite tree in which each vertex has four oriented edges coming out of it, corresponding to

$$a, a^{-1}, b, b^{-1}.$$

A small schematic picture is:



This is only a local sketch; the actual universal cover continues infinitely in every direction and contains no cycles. It is a tree, so it is simply connected.

Each closed reduced word in $a^{\pm 1}, b^{\pm 1}$ corresponds to a loop downstairs in R_2 , while upstairs it becomes a path in the tree. Such a loop is trivial in F_2 exactly when the lifted path begins and ends at the same vertex without moving, equivalently when the word reduces to the empty word.

This makes the universal cover of R_2 a geometric model for reduced words in the free group F_2 .

4.21 Covering graphs and subgroups of free groups

Because

$$\pi_1(R_n) \cong F_n,$$

connected covering graphs of the rose R_n correspond to subgroups of F_n .

This is the graph-theoretic form of the covering-space classification theorem. It is especially powerful because the covering can often be drawn explicitly.

If

$$p : \tilde{\Gamma} \rightarrow R_n$$

is a connected covering graph, then

$$p_*\pi_1(\tilde{\Gamma}) \leq \pi_1(R_n) \cong F_n$$

is a subgroup of the free group F_n . Since $\tilde{\Gamma}$ is itself a graph, its fundamental group is again free. So one gets a topological proof that subgroups of free groups are free.

This is the geometric content of the Nielsen–Schreier theorem.

4.22 A finite-sheeted covering graph of R_2

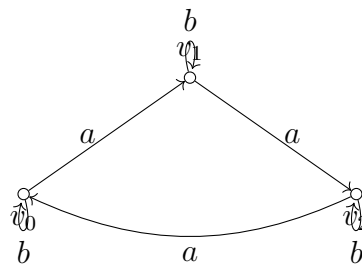
Here is a concrete connected 3-sheeted covering of the rose R_2 with loops labeled a and b . Let the vertices upstairs be v_0, v_1, v_2 . Let the a -edges form a 3-cycle

$$v_0 \rightarrow v_1 \rightarrow v_2 \rightarrow v_0,$$

and let each vertex have a b -loop at itself.

Then every vertex has exactly one outgoing a -edge and one outgoing b -edge, matching the local structure downstairs. So this is a covering graph of R_2 .

A picture is:



Now count vertices and edges:

$$V = 3, \quad E = 6.$$

Hence

$$\text{rank } \pi_1(\tilde{\Gamma}) = E - V + 1 = 6 - 3 + 1 = 4.$$

So this covering graph corresponds to a subgroup of index 3 in F_2 that is isomorphic to F_4 . This is a concrete picture of the rank formula

$$\text{rank}(H) = 1 + d(r - 1)$$

with $r = 2$ and $d = 3$.

4.23 A covering graph of the circle corresponding to $n\mathbb{Z}$

Another basic example is the n -sheeted covering

$$S^1 \rightarrow S^1, \quad z \mapsto z^n.$$

In graph language, this is a cycle with n vertices and n edges mapping onto one loop-edge downstairs.

The corresponding subgroup of

$$\pi_1(S^1) \cong \mathbb{Z}$$

is

$$n\mathbb{Z}.$$

So finite-sheeted covering graphs of the circle correspond exactly to subgroups $n\mathbb{Z} \leq \mathbb{Z}$.

4.24 Deck transformations of covering graphs

If

$$p : \tilde{\Gamma} \rightarrow \Gamma$$

is a covering graph, a *deck transformation* is a homeomorphism

$$h : \tilde{\Gamma} \rightarrow \tilde{\Gamma}$$

such that

$$p \circ h = p.$$

For the universal cover of the circle, the deck group is $\mathbb{Z}$, acting by translations on the infinite line.

For the universal cover of R_2 , the deck group is

$$\text{Deck}(\tilde{R}_2/R_2) \cong \pi_1(R_2) \cong F_2.$$

So the free group acts on its universal covering tree by graph automorphisms.

This point of view is extremely important in geometric group theory: a group can often be understood by how it acts on a graph or tree.

4.25 Why covering graphs are so useful

Covering graphs make several abstract facts concrete.

- They turn subgroups of free groups into pictures.
- They give geometric proofs that subgroups of free groups are free.
- They make the rank formula visible by counting vertices and edges.
- They let words in generators be interpreted as paths in a graph.
- They prepare the way for Bass–Serre theory, where groups are studied via actions on trees.

Thus, whenever free groups appear, it is very natural to draw graphs and covering graphs. They are not just illustrations: they are the underlying geometry of the algebra.

4.26 A short summary

The main facts about graphs and covering graphs used in Problems 7–9 are:

- A connected graph has free fundamental group.
- If a finite connected graph has V vertices and E edges, then

$$\pi_1(\Gamma) \cong F_{E-V+1}.$$

- Trees are contractible and have trivial fundamental group.
- The universal cover of a graph is a tree.
- Connected covering graphs of R_n correspond to subgroups of F_n .

- Finite-sheeted covers of R_n give finite-index subgroups of F_n .
- Counting vertices and edges in the covering graph recovers the subgroup rank.

This is the graph-theoretic backbone of the theory of free groups.

5 Collapsing the equator of S^2 : why $S^2/S^1 \cong S^2 \vee S^2$

5.1 Statement of the quotient result

Let $S^1 \subset S^2$ be the equator. Then

$$S^2/S^1 \cong S^2 \vee S^2.$$

Here S^2/S^1 means that the entire equator is collapsed to a single point, and

$$S^2 \vee S^2$$

denotes the wedge sum of two 2-spheres, that is, two spheres with one chosen point on each identified.

Equivalently: if one takes the standard sphere, splits it into upper and lower hemispheres, and then shrinks the equator to a point, the result is two spheres touching at exactly one point.

5.2 Decomposing the sphere into two hemispheres

Write the sphere as

$$S^2 = D_+^2 \cup D_-^2, \quad D_+^2 \cap D_-^2 = S^1,$$

where:

- D_+^2 is the upper hemisphere,
- D_-^2 is the lower hemisphere,
- their common boundary S^1 is the equator.

So topologically the sphere is obtained by gluing two closed disks along their boundary circle.

5.3 The key fact: $D^2/\partial D^2 \cong S^2$

The main ingredient is the standard quotient

$$D^2/\partial D^2 \cong S^2.$$

This means: if one takes a closed disk and collapses its entire boundary circle to one point, the result is homeomorphic to a 2-sphere.

Geometrically, this is very natural. A disk already looks like one hemisphere of a sphere. The boundary circle is what would become the equator. If that boundary is crushed to one point, the disk closes up into a sphere.

Thus each hemisphere of S^2 , after collapsing its boundary equator, becomes a copy of S^2 .

5.4 Applying this to the upper and lower hemispheres

Now collapse the equator $S^1 \subset S^2$ to one point.

Since the equator is the boundary of both hemispheres, this means:

$$D_+^2/\partial D_+^2 \cong S^2, \quad D_-^2/\partial D_-^2 \cong S^2.$$

So:

- the upper hemisphere becomes one sphere,
- the lower hemisphere becomes another sphere.

But both hemisphere boundaries are identified to the *same* collapsed point, because the equator is collapsed globally inside S^2 .

Therefore the two resulting spheres meet at exactly one point. This is precisely the wedge sum:

$$S^2 \vee S^2.$$

Hence

$$S^2/S^1 \cong S^2 \vee S^2.$$

5.5 Formal quotient description

Since

$$S^2 = D_+^2 \cup_{S^1} D_-^2,$$

collapsing the common equator gives

$$S^2/S^1 = (D_+^2 \cup_{S^1} D_-^2)/S^1.$$

After the collapse, each hemisphere has its boundary identified to the same point, so the quotient becomes

$$S^2/S^1 \cong (D_+^2/\partial D_+^2) \vee (D_-^2/\partial D_-^2).$$

Using

$$D^2/\partial D^2 \cong S^2,$$

we obtain

$$S^2/S^1 \cong S^2 \vee S^2.$$

5.6 Why this is a wedge sum

Recall that the wedge sum $X \vee Y$ is defined by choosing one point in X , one point in Y , and identifying them.

That is exactly what happens here. Each hemisphere becomes a sphere, and both are attached at the single point that comes from the collapsed equator.

So the quotient is:

- not two disjoint spheres,
- not one sphere again,
- not a torus,
- but exactly two spheres joined at one point.

Hence it is the wedge sum

$$S^2 \vee S^2.$$

5.7 Geometric intuition

Before taking the quotient:

- there is one sphere,
- it is divided into two hemispheres,
- the hemispheres meet along the equator circle.

After collapsing the equator:

- the equator shrinks to one point,
- the northern hemisphere becomes a sphere,
- the southern hemisphere becomes another sphere,
- the two spheres touch at the collapsed equator point.

So visually the process is:

$$\text{sphere cut along the equator} \longrightarrow \text{two spheres joined at one point.}$$

5.8 A homology check

This conclusion is also confirmed by homology.

For the wedge sum $S^2 \vee S^2$, one has

$$\widetilde{H}_2(S^2 \vee S^2) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

This reflects the fact that there are two 2-dimensional sphere pieces.

That is exactly what one expects after collapsing the equator:

- one 2-dimensional class comes from the upper hemisphere,

- one 2-dimensional class comes from the lower hemisphere.

So the algebra agrees with the geometry.

5.9 Compact conclusion

$$\boxed{S^2/S^1 \cong S^2 \vee S^2.}$$

The reason is that

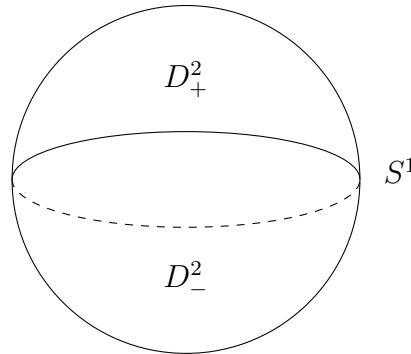
$$S^2 = D_+^2 \cup_{S^1} D_-^2,$$

and collapsing the common equator turns each disk into a sphere:

$$D^2/\partial D^2 \cong S^2.$$

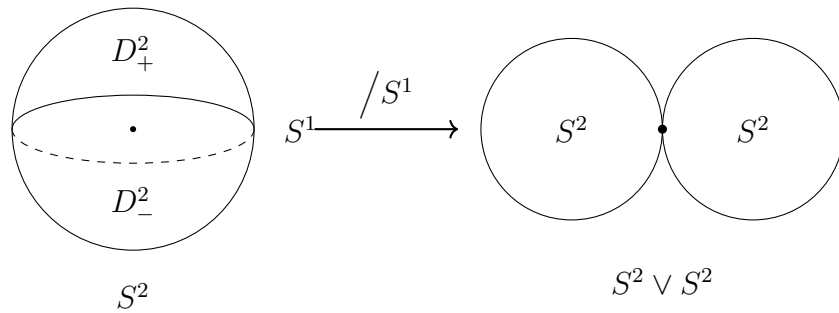
Since both boundaries are collapsed to the same point, the two resulting spheres are glued at one point, which is exactly the wedge sum.

5.10 Diagram: sphere split into hemispheres

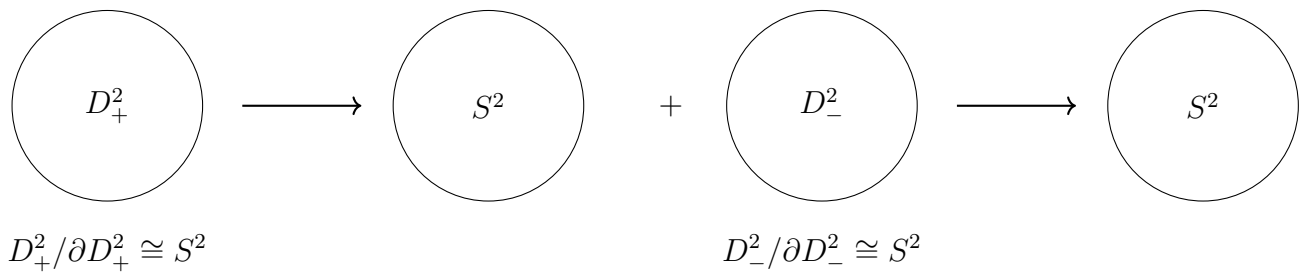


$$S^2 = D_+^2 \cup D_-^2, \quad D_+^2 \cap D_-^2 = S^1$$

5.11 Diagram: collapsing the equator



5.12 Diagram: each hemisphere becomes a sphere



5.13 A short proposition-and-proof version

Proposition 5.1. *Let $S^1 \subset S^2$ be the equator. Then*

$$S^2 / S^1 \cong S^2 \vee S^2.$$

Proof. Write

$$S^2 = D_+^2 \cup D_-^2, \quad D_+^2 \cap D_-^2 = S^1.$$

Collapsing the equator S^1 to a point collapses the boundary of each hemisphere to that same point. Therefore

$$S^2 / S^1 \cong (D_+^2 / \partial D_+^2) \vee (D_-^2 / \partial D_-^2).$$

Since

$$D^2 / \partial D^2 \cong S^2,$$

it follows that

$$S^2 / S^1 \cong S^2 \vee S^2.$$

□

5.14 Optional remark: general pattern

This is a special case of a general phenomenon: when a space is split into two pieces along a common boundary and that entire common boundary is collapsed to one point, the quotient often becomes a wedge of the corresponding quotient pieces.

Here the common boundary is the equator S^1 , the pieces are the two hemispheres D_+^2 and D_-^2 , and each quotient

$$D^2 / \partial D^2$$

is a sphere, so the result is

$$S^2 \vee S^2.$$

6 Problem 10 Hopf link versus the two-component unlink in S^3

Problem statement. Consider the following configurations of pairs of circles in S^3 (we have drawn them in $\mathbb{R}^3$; add a point at infinity). By computing the fundamental groups

of the complements of the circles, show there is no homeomorphism of S^3 taking one configuration to the other.

Solution. Let L_H denote the left-hand configuration and L_U the right-hand configuration.

The left-hand configuration is the *Hopf link*: the two circles are linked once. The right-hand configuration is the *two-component unlink*: the two circles are disjoint and unlinked.

We compute the fundamental groups of their complements.

1. The complement of the unlink

Let

$$X_U = S^3 \setminus L_U.$$

It is standard that removing the circles themselves or removing disjoint open tubular neighborhoods of the circles gives spaces with the same homotopy type. So we may work with the *link exterior*

$$E_U = S^3 \setminus \text{int}(N(L_U)).$$

Since L_U is the unlink of two components, $N(L_U)$ is the disjoint union of two solid tori embedded in S^3 in an unlinked way. The complement E_U is therefore a handlebody of genus 2. A genus 2 handlebody deformation retracts onto a wedge of two circles, so

$$\pi_1(X_U) \cong \pi_1(E_U) \cong F_2,$$

the free group on two generators.

Thus

$$\pi_1(S^3 \setminus L_U) \cong F_2.$$

2. The complement of the Hopf link

Now let

$$X_H = S^3 \setminus L_H.$$

Take one component K_1 of the Hopf link. The complement of an unknot in S^3 is a solid torus:

$$S^3 \setminus K_1 \cong S^1 \times D^2.$$

Inside this solid torus, the second component K_2 of the Hopf link is isotopic to the core circle

$$S^1 \times \{0\}.$$

Hence

$$X_H \cong (S^1 \times D^2) \setminus (S^1 \times \{0\}) \cong S^1 \times (D^2 \setminus \{0\}).$$

But

$$D^2 \setminus \{0\} \cong S^1 \times (0, 1),$$

so

$$X_H \cong S^1 \times S^1 \times (0, 1) \cong T^2 \times (0, 1).$$

Therefore

$$\pi_1(X_H) \cong \pi_1(T^2) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

Thus

$$\pi_1(S^3 \setminus L_H) \cong \mathbb{Z}^2.$$

3. Comparison

We have shown:

$$\pi_1(S^3 \setminus L_H) \cong \mathbb{Z}^2, \quad \pi_1(S^3 \setminus L_U) \cong F_2.$$

These groups are not isomorphic, since $\mathbb{Z}^2$ is abelian while F_2 is nonabelian.

If there were a homeomorphism of S^3 taking one configuration to the other, it would induce a homeomorphism between the complements, hence an isomorphism of their fundamental groups. Since no such isomorphism exists, no such homeomorphism exists.

Therefore the two configurations are not equivalent under any homeomorphism of S^3 .

7 Problem 11 Trees, quotient graphs, coverings, and free groups

Problem statement. A graph G is a space obtained by starting with a set $E(G)$ of copies of the interval I and an equivalence relation $\sim$ on $E(G) \times \{0, 1\}$, and forming the quotient space of $E(G) \times I$ by the minimal equivalence relation containing $\sim$. (That is, it is a space obtained from a set of copies of I by gluing their ends together.) The *vertices* are the equivalence classes represented by the ends of the intervals.

- (i) A tree is a simply-connected graph. A star is a tree with a vertex x_0 such that one end of each edge is attached to x_0 . A leaf of a tree is a vertex attached to only one edge. Prove that every tree is homotopy equivalent to a star, relative to its leaves.
- (ii) If $T \subset G$ is a tree, show that the quotient map $G \rightarrow G/T$ is a homotopy equivalence, and that G/T is again a graph. Hence show that every connected graph is homotopy equivalent to a graph with a single vertex. [*You should assume that every graph has a maximal tree*]
- (iii) Show that the fundamental group of a graph with one vertex, based at the vertex, is a free group with one generator for each edge of the graph. Hence show that any free group occurs as the fundamental group of some graph. (*We have not required that any graph have one vertex*.)
- (iv) Show that a covering space of a graph is again a graph, and deduce that a subgroup of a free group is again a free group.

Solution.

(i) Every tree is homotopy equivalent to a star, relative to its leaves

Let T be a tree.

Consider the subgraph $C \subset T$ consisting of all edges that are *not* incident to a leaf, together with their endpoints.

If $C = \emptyset$, then every edge of T is incident to a leaf. In that case T is already a star.

Assume $C \neq \emptyset$. Then C is connected: if e_1 and e_2 are two edges of C , the unique path in the tree joining them cannot pass through a leaf, since a leaf cannot lie in the interior of a path joining two other edges. Hence every edge on that path also lies in C .

Thus C is a connected subtree, hence contractible.

Now collapse C to a point. Since C is contractible, the quotient map

$$T \longrightarrow T/C$$

is a homotopy equivalence. Moreover, each component attached to C outside C is just an edge ending in a leaf, so after collapsing C to one point, the quotient T/C is a star. The leaves remain distinct and are fixed during the collapse, so this is a homotopy equivalence *relative to the leaves*.

Therefore every tree is homotopy equivalent to a star, relative to its leaves.

(ii) Collapsing a tree inside a graph

Let $T \subset G$ be a tree.

Since T is contractible, there is a homotopy in T from the identity map to a constant map $T \rightarrow \{x_0\}$, where $x_0 \in T$. Because T is a subcomplex of the graph G , this homotopy extends to a homotopy of all of G . Hence collapsing T to a point does not change the homotopy type: the quotient map

$$q : G \longrightarrow G/T$$

is a homotopy equivalence.

Also, G/T is again a graph: all of the edges and vertices of T are collapsed to one vertex, while every edge of $G \setminus T$ remains an interval whose endpoints are attached to quotient vertices.

Now let G be connected, and choose a maximal tree $T \subset G$. Then G/T is a graph with exactly one vertex, since all vertices of G lie in the maximal tree T . By the first part, G is homotopy equivalent to G/T , which is a graph with a single vertex.

Therefore every connected graph is homotopy equivalent to a graph with one vertex.

(iii) Fundamental group of a one-vertex graph

Let G be a graph with one vertex v . Then every edge begins and ends at v , so topologically

$$G \cong \bigvee_{e \in E(G)} S_e^1,$$

a wedge of circles, one for each edge.

If $E(G)$ is finite, say $E(G) = \{e_1, \dots, e_n\}$, then by van Kampen,

$$\pi_1(G, v) \cong F_n,$$

the free group on n generators, one generator for each loop e_i .

If $E(G)$ is infinite, any loop $\alpha : S^1 \rightarrow G$ has compact image, so it meets only finitely many open edges. Hence α is contained in a finite subgraph

$$G_F = \bigvee_{e \in F} S_e^1$$

for some finite subset $F \subset E(G)$. Therefore every element of $\pi_1(G, v)$ lies in the image of $\pi_1(G_F, v) \cong F(F)$, and passing to the direct limit over finite F , we obtain

$$\pi_1(G, v) \cong F(E(G)),$$

the free group on the set of edges.

Thus the fundamental group of a graph with one vertex is a free group with one generator for each edge.

Conversely, given any free group $F(S)$ on a set S , take a graph with one vertex and one loop for each element of S . Its fundamental group is exactly $F(S)$.

Hence every free group occurs as the fundamental group of some graph.

(iv) Coverings of graphs and subgroups of free groups

Let

$$p : \tilde{G} \longrightarrow G$$

be a covering map, where G is a graph.

Each open edge $e^\circ \subset G$ is homeomorphic to an open interval. Since coverings are local homeomorphisms, $p^{-1}(e^\circ)$ is a disjoint union of open intervals. The preimage of the vertex set consists of points, and a neighborhood of each such point is mapped homeomorphically onto a star-neighborhood of a vertex in G . Therefore $\tilde{G}$ is again obtained by gluing intervals along endpoints; that is, $\tilde{G}$ is again a graph.

Now let F be a free group, and let $H \leq F$ be a subgroup. By part (iii), choose a connected graph G with

$$\pi_1(G, v) \cong F.$$

By covering space theory, the subgroup H corresponds to a connected covering space

$$p : (\tilde{G}, \tilde{v}) \rightarrow (G, v)$$

such that

$$p_*(\pi_1(\tilde{G}, \tilde{v})) = H.$$

Since $\tilde{G}$ is again a graph, parts (ii) and (iii) imply that

$$\pi_1(\tilde{G}, \tilde{v})$$

is a free group. Therefore H is free.

This proves that every subgroup of a free group is again a free group.

8 Hopf link versus the two-component unlink in S^3

8.1 Definitions and background

Links and knots. A *link* in S^3 is a finite disjoint union of embedded circles

$$L = K_1 \sqcup K_2 \sqcup \cdots \sqcup K_m \subset S^3.$$

If $m = 1$, then L is called a *knot*.

Complement and exterior. Given a link $L \subset S^3$, its *complement* is

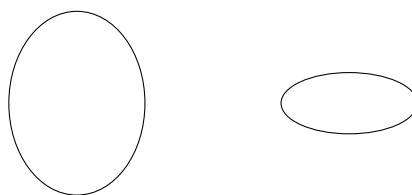
$$S^3 \setminus L.$$

Often one removes instead a small open tubular neighborhood $N(L)$, and studies the *link exterior*

$$E(L) = S^3 \setminus \text{int}(N(L)).$$

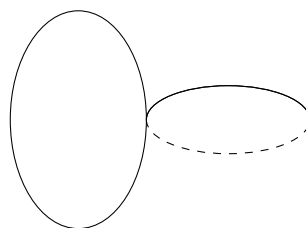
The spaces $S^3 \setminus L$ and $E(L)$ have the same homotopy type, so they have the same fundamental group.

The two-component unlink. The *two-component unlink* is a pair of disjoint circles in S^3 which are not linked.



Two-component unlink

The Hopf link. The *Hopf link* is the simplest nontrivial two-component link: each component links the other once.



Hopf link

Why complements matter. If a homeomorphism $h : S^3 \rightarrow S^3$ carries one link L_1 onto another link L_2 , then it restricts to a homeomorphism

$$S^3 \setminus L_1 \cong S^3 \setminus L_2.$$

Hence their fundamental groups must be isomorphic:

$$\pi_1(S^3 \setminus L_1) \cong \pi_1(S^3 \setminus L_2).$$

Therefore, if the complement groups are not isomorphic, then no homeomorphism of S^3 can carry one link to the other.

8.2 Problem statement

Problem. Consider the following configurations of pairs of circles in S^3 (drawn in $\mathbb{R}^3$, with a point at infinity added). By computing the fundamental groups of the complements of the circles, show that there is no homeomorphism of S^3 carrying one configuration to the other.

8.3 Solution

Let L_U denote the unlink, and let L_H denote the Hopf link.

Step 1: the complement of the unlink. Consider

$$X_U = S^3 \setminus L_U.$$

Replacing L_U by a disjoint union of small tubular neighborhoods does not change the homotopy type, so we may study the exterior

$$E_U = S^3 \setminus \text{int}(N(L_U)).$$

Since L_U is the unlink of two components, $N(L_U)$ is the disjoint union of two unknotted solid tori in S^3 , placed in an unlinked way. Their complement is a handlebody of genus 2.

A genus 2 handlebody deformation retracts onto a wedge of two circles:

$$E_U \simeq S^1 \vee S^1.$$

Therefore

$$\pi_1(S^3 \setminus L_U) \cong \pi_1(E_U) \cong \pi_1(S^1 \vee S^1) \cong F_2,$$

the free group on two generators.



$$S^1 \vee S^1$$

So the unlink complement has nonabelian fundamental group:

$$\pi_1(S^3 \setminus L_U) \cong F_2.$$

Step 2: the complement of the Hopf link. Now consider

$$X_H = S^3 \setminus L_H.$$

Remove one component K_1 first. Since K_1 is an unknot, its complement is a solid torus:

$$S^3 \setminus K_1 \cong S^1 \times D^2.$$

Inside this solid torus, the second component K_2 of the Hopf link is isotopic to the core circle $S^1 \times \{0\}$. Hence

$$X_H \cong (S^1 \times D^2) \setminus (S^1 \times \{0\}) \cong S^1 \times (D^2 \setminus \{0\}).$$

Since

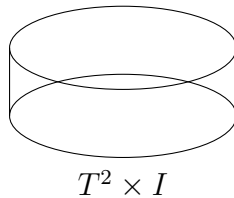
$$D^2 \setminus \{0\} \cong S^1 \times (0, 1),$$

we obtain

$$X_H \cong S^1 \times S^1 \times (0, 1) \cong T^2 \times (0, 1).$$

Therefore

$$\pi_1(S^3 \setminus L_H) \cong \pi_1(T^2) \cong \mathbb{Z} \oplus \mathbb{Z}.$$



So the Hopf link complement has abelian fundamental group:

$$\pi_1(S^3 \setminus L_H) \cong \mathbb{Z}^2.$$

Step 3: compare the groups. We have shown

$$\pi_1(S^3 \setminus L_U) \cong F_2, \quad \pi_1(S^3 \setminus L_H) \cong \mathbb{Z}^2.$$

These groups are not isomorphic, because F_2 is nonabelian while $\mathbb{Z}^2$ is abelian.

Hence the complements are not homeomorphic. Therefore no homeomorphism of S^3 can carry the unlink to the Hopf link.

Conclusion. There is no homeomorphism of S^3 taking one of the two configurations to the other.

8.4 Further examples and remarks

Example 1: the unknot. The complement of the unknot has the homotopy type of a circle:

$$S^3 \setminus (\text{unknot}) \simeq S^1, \quad \pi_1 \cong \mathbb{Z}.$$

Example 2: the n -component unlink. The complement of the n -component unlink has fundamental group

$$\pi_1 \cong F_n.$$

Example 3: the Hopf link. The Hopf link is the first example where the complement looks torus-like:

$$S^3 \setminus (\text{Hopf link}) \cong T^2 \times (0, 1), \quad \pi_1 \cong \mathbb{Z}^2.$$

Linking number. Another classical invariant is the linking number:

$$\text{lk}(K_1, K_2).$$

For the unlink one has $\text{lk} = 0$, while for the Hopf link one has $\text{lk} = \pm 1$. The problem, however, asks us to distinguish the two configurations via the fundamental groups of their complements.

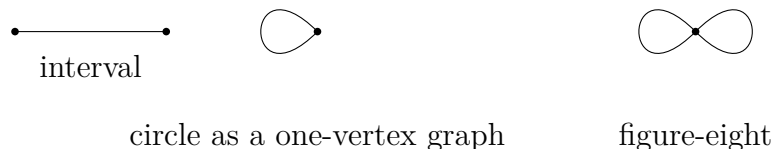
9 Graphs, trees, coverings, and free groups

9.1 Definitions and examples

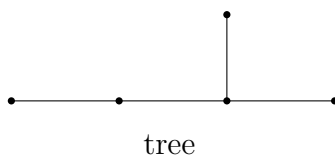
Topological graph. A *graph* is a topological space obtained by taking a collection of copies of the interval $I = [0, 1]$ and gluing endpoints together. Equivalently, a graph is a 1-dimensional CW-complex.

Vertices and edges. The images of the endpoints are the *vertices*, and the images of the intervals are the *edges*.

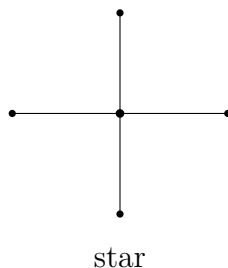
Examples.



Tree. A *tree* is a connected graph with no cycles. In the present context, this is equivalent to a simply connected graph.



Star. A *star* is a tree with a central vertex x_0 such that each edge has one endpoint at x_0 .



Leaf. A *leaf* of a tree is a vertex attached to exactly one edge.

Maximal tree. If G is a connected graph, a *maximal tree* $T \subset G$ is a subtree containing all vertices of G and as many edges as possible without creating a cycle. This is the same as a spanning tree.

9.2 Problem statement

Problem. A graph G is a space obtained by starting with a set $E(G)$ of copies of the interval I and an equivalence relation $\sim$ on $E(G) \times \{0, 1\}$, and forming the quotient space of $E(G) \times I$ by the minimal equivalence relation containing $\sim$. That is, it is a space obtained from a set of copies of I by gluing their ends together. The vertices are the equivalence classes represented by the ends of the intervals.

- (i) A tree is a simply-connected graph. A star is a tree with a vertex x_0 such that one end of each edge is attached to x_0 . A leaf of a tree is a vertex attached to only one edge. Prove that every tree is homotopy equivalent to a star, relative to its leaves.
- (ii) If $T \subset G$ is a tree, show that the quotient map $G \rightarrow G/T$ is a homotopy equivalence, and that G/T is again a graph. Hence show that every connected graph is homotopy equivalent to a graph with a single vertex. [You should assume that every graph has a maximal tree.]
- (iii) Show that the fundamental group of a graph with one vertex, based at the vertex, is a free group with one generator for each edge of the graph. Hence show that any free group occurs as the fundamental group of some graph. (We have not required that any graph have finitely many edges.)
- (iv) Show that a covering space of a graph is again a graph, and deduce that a subgroup of a free group is again a free group.

9.3 Solution

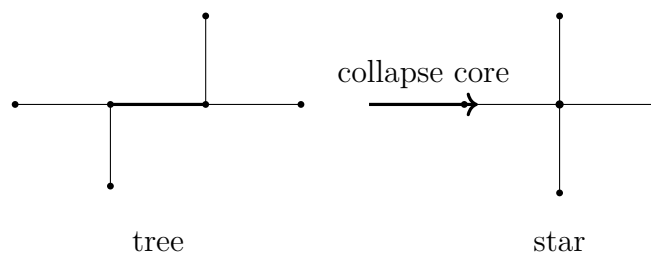
(i) Every tree is homotopy equivalent to a star, relative to its leaves.

Let T be a tree. If T is already a star, there is nothing to prove. Otherwise, define $C \subset T$ to be the union of all edges which are not incident to a leaf, together with their endpoints. Intuitively, C is the internal core of the tree.

We claim that C is connected. Indeed, if e_1 and e_2 are two edges of C , then the unique path in the tree joining a point of e_1 to a point of e_2 cannot pass through a leaf in its interior, since a leaf has valence 1. Hence all edges on that path also belong to C . Thus C is connected.

Since C is a connected subgraph of a tree, it is again a tree, hence contractible. Now collapse C to a single point. Because C is contractible, this collapse does not change the homotopy type. After the collapse, every remaining edge has one endpoint at the new central point and the other at a leaf. Thus the quotient is a star.

Moreover, the leaves are not identified with one another and may be kept fixed throughout the deformation, so the homotopy equivalence is relative to the leaves.



(ii) Collapsing a tree inside a graph.

Let $T \subset G$ be a tree. Since T is contractible, collapsing T to a point should not change the homotopy type. More precisely, the quotient map

$$q : G \rightarrow G/T$$

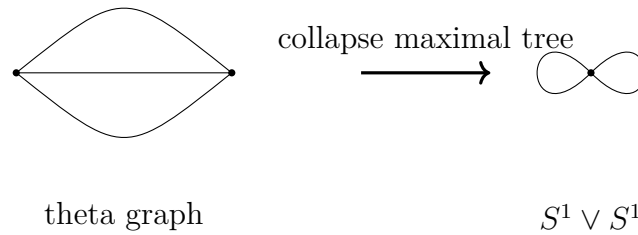
is a homotopy equivalence.

The quotient G/T is again a graph: one replaces the entire subtree T by a single vertex, and every edge outside T remains an interval attached at its endpoints to quotient vertices.

Now suppose that G is connected, and let $T \subset G$ be a maximal tree. Since T contains every vertex of G , collapsing T to a point identifies all vertices of G to one single vertex. Therefore G/T is a graph with exactly one vertex.

Hence every connected graph is homotopy equivalent to a graph with a single vertex.

Example: the theta graph. Consider the graph with two vertices joined by three distinct edges:



So the theta graph is homotopy equivalent to a wedge of two circles.

(iii) Fundamental group of a graph with one vertex.

Suppose G has one vertex v . Then each edge begins and ends at v , so topologically

$$G \cong \bigvee_{e \in E(G)} S_e^1,$$

a wedge of circles, one for each edge.

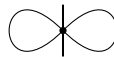
If G has n edges, then

$$G \cong \bigvee_{i=1}^n S^1,$$

and by van Kampen's theorem,

$$\pi_1(G, v) \cong F_n,$$

the free group on n generators.



rose with 3 petals

Thus each edge contributes one free generator.

If the graph has infinitely many edges, the same conclusion still holds. Any loop $\alpha : S^1 \rightarrow G$ has compact image, so it meets only finitely many edges; therefore α is contained in a finite wedge of circles. Passing to the direct limit over finite subgraphs, one obtains that $\pi_1(G, v)$ is the free group on the whole edge set.

Conversely, given any free group $F(S)$ on a set S , take one vertex and attach one loop for each element of S . The resulting graph has fundamental group $F(S)$.

So every free group occurs as the fundamental group of some graph.

Rank formula for finite connected graphs. If G is a finite connected graph with V vertices and E edges, then

$$\text{rank } \pi_1(G) = E - V + 1.$$

Indeed, a maximal tree has $V - 1$ edges, so the number of edges not in the maximal tree is

$$E - (V - 1) = E - V + 1,$$

and each such extra edge contributes one free generator.

Examples.

- A triangle has $V = 3$, $E = 3$, so

$$\text{rank } \pi_1 = 3 - 3 + 1 = 1,$$

hence $\pi_1 \cong \mathbb{Z}$.

- A figure-eight has $V = 1$, $E = 2$, so

$$\text{rank } \pi_1 = 2 - 1 + 1 = 2,$$

hence $\pi_1 \cong F_2$.

- The theta graph has $V = 2$, $E = 3$, so

$$\text{rank } \pi_1 = 3 - 2 + 1 = 2,$$

hence $\pi_1 \cong F_2$.

(iv) Coverings of graphs and subgroups of free groups.

Let

$$p : \tilde{G} \rightarrow G$$

be a covering map, where G is a graph.

A graph is locally very simple:

- near a point in the interior of an edge, it looks like an open interval;
- near a vertex, it looks like a star-shaped neighborhood.

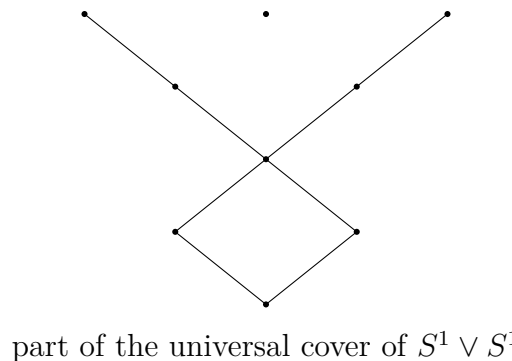
Since a covering map is a local homeomorphism, every point of $\tilde{G}$ has a neighborhood homeomorphic to one of these same local models. Thus $\tilde{G}$ is again obtained by gluing intervals at endpoints, so $\tilde{G}$ is again a graph.

Example: universal cover of S^1 . The universal covering of the circle is the real line:

$$\mathbb{R} \rightarrow S^1.$$


$\mathbb{R}$

Example: universal cover of $S^1 \vee S^1$. The universal cover of the figure-eight graph is an infinite tree.



Now let F be a free group, and let $H \leq F$ be a subgroup. By part (iii), there exists a connected graph G with

$$\pi_1(G, v) \cong F.$$

By covering space theory, the subgroup H corresponds to a connected covering

$$p : (\tilde{G}, \tilde{v}) \rightarrow (G, v)$$

such that

$$p_* (\pi_1(\tilde{G}, \tilde{v})) = H.$$

But $\tilde{G}$ is again a graph, so by part (iii), its fundamental group is free. Therefore H is free. This proves the Nielsen–Schreier theorem:

Theorem. Every subgroup of a free group is free.

9.4 Conceptual summary

These two problems illustrate two central ideas of algebraic topology.

- In knot and link theory, one studies an embedding in S^3 by studying the topology of its complement. In Problem 10, the complement groups distinguish the Hopf link from the unlink.
- In the topology of graphs, one sees free groups geometrically. A connected graph is homotopy equivalent to a bouquet of circles, and its fundamental group is therefore free. Covering spaces of graphs then explain why subgroups of free groups are again free.

Thus, in both problems, the fundamental group acts as a bridge between geometry and algebra.

10 The fundamental group: definitions, theory, and examples

10.1 Basic idea and motivation

The fundamental group is one of the central invariants of algebraic topology. It measures, in algebraic form, the existence of nontrivial loops in a space.

Very roughly:

- if every loop can be continuously contracted to a point, the space is simply connected and its fundamental group is trivial;
- if some loops cannot be contracted, the fundamental group records how such loops wind around holes.

The key idea is to replace geometric information about loops by algebraic information in a group.

10.2 Paths, loops, and homotopies

Definition (path). Let X be a topological space. A *path* in X is a continuous map

$$\gamma : [0, 1] \rightarrow X.$$

The point $\gamma(0)$ is the initial point, and $\gamma(1)$ is the terminal point.

Definition (loop). A *loop* in X based at $x_0 \in X$ is a path

$$\gamma : [0, 1] \rightarrow X$$

such that

$$\gamma(0) = \gamma(1) = x_0.$$

Definition (homotopy of paths). Two paths $\gamma, \eta : [0, 1] \rightarrow X$ with the same endpoints are *path-homotopic relative to endpoints* if there exists a continuous map

$$H : [0, 1] \times [0, 1] \rightarrow X$$

such that

$$H(s, 0) = \gamma(s), \quad H(s, 1) = \eta(s),$$

and

$$H(0, t) = \gamma(0) = \eta(0), \quad H(1, t) = \gamma(1) = \eta(1)$$

for all $t \in [0, 1]$.

Thus a homotopy deforms one path into the other while keeping the endpoints fixed.

Definition (homotopy of loops). Two loops based at x_0 are *homotopic as based loops* if they are path-homotopic relative to the basepoint.

10.3 Definition of the fundamental group

Fix a point $x_0 \in X$.

Definition. The *fundamental group* of X at x_0 , denoted

$$\pi_1(X, x_0),$$

is the set of homotopy classes of loops based at x_0 , with group operation given by concatenation of loops.

If $[\alpha]$ and $[\beta]$ are homotopy classes of loops based at x_0 , then their product is

$$[\alpha] \cdot [\beta] = [\alpha * \beta],$$

where

$$(\alpha * \beta)(t) = \begin{cases} \alpha(2t), & 0 \leq t \leq \frac{1}{2}, \\ \beta(2t - 1), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

Identity element. The identity is the class of the constant loop

$$c_{x_0}(t) = x_0.$$

Inverse. If α is a loop, then the inverse of $[\alpha]$ is represented by the reversed loop

$$\bar{\alpha}(t) = \alpha(1 - t).$$

Remark. Strictly speaking, concatenation of loops is not associative as literal maps, but it is associative up to homotopy. That is enough to make $\pi_1(X, x_0)$ into a group.

10.4 Why this is a topological invariant

A continuous map

$$f : (X, x_0) \rightarrow (Y, y_0)$$

with $f(x_0) = y_0$ sends a loop α in X based at x_0 to the loop $f \circ \alpha$ in Y based at y_0 . This induces a group homomorphism

$$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0), \quad f_*([\alpha]) = [f \circ \alpha].$$

Thus the fundamental group is functorial.

In particular, if $f : X \rightarrow Y$ is a homeomorphism, then

$$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0))$$

is an isomorphism. Therefore homeomorphic spaces have isomorphic fundamental groups. More generally, homotopy equivalent spaces have isomorphic fundamental groups.

10.5 Basepoint issues

The definition uses a basepoint x_0 , so in general $\pi_1(X, x_0)$ depends on that choice.

However, if X is path-connected and $x_0, x_1 \in X$, then any path from x_0 to x_1 gives an isomorphism

$$\pi_1(X, x_0) \cong \pi_1(X, x_1).$$

This is not always canonical, but for path-connected spaces the fundamental group is essentially independent of the basepoint up to isomorphism.

Because of this, one often simply writes $\pi_1(X)$ when the basepoint is not important.

10.6 Simply connected spaces

Definition. A space X is called *simply connected* if

- X is path-connected, and

- $\pi_1(X, x_0)$ is trivial for some (hence every) basepoint $x_0 \in X$.

Equivalently, every loop in X can be contracted to a point.

Examples of simply connected spaces.

- $\mathbb{R}^n$ for every $n \geq 1$,
- any convex subset of $\mathbb{R}^n$,
- the disk D^n ,
- the sphere S^n for $n \geq 2$,
- every tree.

10.7 First basic examples

Example 1: a point. If $X = \{*\}$ is a single point, then every loop is constant, so

$$\pi_1(X) \cong \{e\}.$$

Example 2: Euclidean space. Since $\mathbb{R}^n$ is contractible,

$$\pi_1(\mathbb{R}^n) \cong \{e\}.$$

Example 3: the circle. The most important first example is

$$\pi_1(S^1) \cong \mathbb{Z}.$$

The integer records the winding number of a loop around the circle.

A loop that goes once counterclockwise represents 1, once clockwise represents -1 , twice counterclockwise represents 2, and so on.

Example 4: the torus. The torus is

$$T^2 = S^1 \times S^1.$$

Hence

$$\pi_1(T^2) \cong \pi_1(S^1) \times \pi_1(S^1) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

One generator comes from going once around one circular direction, and the other comes from going once around the other.

Example 5: the sphere. For $n \geq 2$,

$$\pi_1(S^n) \cong \{e\}.$$

In particular,

$$\pi_1(S^2) \cong \{e\}.$$

Example 6: punctured plane. The punctured plane

$$\mathbb{R}^2 \setminus \{0\}$$

deformation retracts onto a circle, so

$$\pi_1(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{Z}.$$

Example 7: punctured three-space. The space

$$\mathbb{R}^3 \setminus \{0\}$$

deformation retracts onto S^2 , so

$$\pi_1(\mathbb{R}^3 \setminus \{0\}) \cong \pi_1(S^2) \cong \{e\}.$$

Thus removing one point from $\mathbb{R}^2$ changes the fundamental group, but removing one point from $\mathbb{R}^3$ does not create a nontrivial fundamental group.

10.8 Deformation retractions and homotopy equivalence

The fundamental group is especially useful together with deformation retractions.

Definition. A subspace $A \subset X$ is a *deformation retract* of X if there exists a continuous map

$$H : X \times [0, 1] \rightarrow X$$

such that

$$H(x, 0) = x, \quad H(x, 1) \in A, \quad H(a, t) = a \text{ for all } a \in A.$$

Then X and A have the same homotopy type, so

$$\pi_1(X) \cong \pi_1(A).$$

Examples.

- $\mathbb{R}^2 \setminus \{0\}$ deformation retracts onto S^1 .
- A solid torus deformation retracts onto its core circle.
- A genus 2 handlebody deformation retracts onto $S^1 \vee S^1$.
- A graph with one vertex and n edges is already a wedge of n circles.

10.9 Wedge sums and free groups

The wedge sum of pointed spaces is formed by identifying their basepoints.

Definition. If (X, x_0) and (Y, y_0) are pointed spaces, then their *wedge sum* is

$$X \vee Y = (X \sqcup Y) / (x_0 \sim y_0).$$

For circles, the wedge

$$S^1 \vee S^1$$

is the figure-eight graph.

A fundamental theorem from van Kampen's theorem says:

$$\pi_1(S^1 \vee S^1) \cong F_2,$$

the free group on two generators.

More generally,

$$\pi_1\left(\bigvee_{i=1}^n S^1\right) \cong F_n.$$

Thus free groups arise naturally as fundamental groups of graphs.

10.10 Van Kampen's theorem

One of the most important tools for computing fundamental groups is van Kampen's theorem.

Informal statement. If a space X is the union of two path-connected open sets U and V , and $U \cap V$ is also path-connected, then $\pi_1(X)$ can be computed from $\pi_1(U)$, $\pi_1(V)$, and $\pi_1(U \cap V)$.

In practice, van Kampen allows us to:

- compute π_1 of wedge sums,
- compute π_1 of cell complexes,
- derive presentations of groups from spaces built by gluing.

For example, applying van Kampen to the figure-eight yields

$$\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z} = F_2.$$

10.11 Fundamental groups of graphs

Graphs are among the most important spaces for understanding fundamental groups.

Theorem. If G is a connected graph, then $\pi_1(G)$ is a free group.

Indeed, if $T \subset G$ is a maximal tree, then collapsing T produces a graph with one vertex and one loop for each edge not in T . Hence

$$G \simeq \bigvee^r S^1$$

for some r , and therefore

$$\pi_1(G) \cong F_r.$$

If G is finite with V vertices and E edges, then

$$r = E - V + 1.$$

So

$$\pi_1(G) \cong F_{E-V+1}.$$

Examples.

(a) **Tree.** A tree has no cycles, so

$$\pi_1(T) \cong \{e\}.$$

(b) **Triangle.** A triangle graph is homotopy equivalent to a circle, so

$$\pi_1 \cong \mathbb{Z}.$$

(c) **Figure-eight.** A figure-eight graph has one vertex and two loop edges, so

$$\pi_1 \cong F_2.$$

(d) **Theta graph.** A theta graph has two vertices and three edges, hence

$$E - V + 1 = 3 - 2 + 1 = 2,$$

so

$$\pi_1 \cong F_2.$$

10.12 Fundamental groups of products

If X and Y are path-connected spaces, then

$$\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y).$$

This gives many important computations.

Examples.

•

$$\pi_1(S^1 \times S^1) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

•

$$\pi_1(S^1 \times D^2) \cong \mathbb{Z},$$

since D^2 is contractible.

•

$$\pi_1(T^2 \times I) \cong \pi_1(T^2) \cong \mathbb{Z}^2,$$

since the interval I is contractible.

10.13 Fundamental groups of complements

One of the richest sources of examples comes from complements of subsets.

Example 1: complement of a point in the plane.

$$\mathbb{R}^2 \setminus \{0\} \simeq S^1, \quad \pi_1 \cong \mathbb{Z}.$$

Example 2: complement of an unknot in S^3 . The complement of an unknot is a solid torus, so

$$\pi_1(S^3 \setminus \text{unknot}) \cong \mathbb{Z}.$$

Example 3: complement of the two-component unlink. The complement deformation retracts onto $S^1 \vee S^1$, so

$$\pi_1(S^3 \setminus \text{unlink}) \cong F_2.$$

Example 4: complement of the Hopf link. The complement is homeomorphic to $T^2 \times (0, 1)$, so

$$\pi_1(S^3 \setminus \text{Hopf link}) \cong \mathbb{Z}^2.$$

These examples show how the topology of a complement can detect the geometric configuration of embedded objects.

10.14 Group presentations from spaces

Often one describes a fundamental group by generators and relations:

$$\pi_1(X, x_0) \cong \langle \text{generators} \mid \text{relations} \rangle.$$

Examples.

(a) **Circle.**

$$\pi_1(S^1) \cong \langle a \rangle \cong \mathbb{Z}.$$

(b) **Figure-eight.**

$$\pi_1(S^1 \vee S^1) \cong \langle a, b \rangle = F_2.$$

(c) **Torus.**

$$\pi_1(T^2) \cong \langle a, b \mid ab = ba \rangle \cong \mathbb{Z}^2.$$

This makes the contrast between F_2 and $\mathbb{Z}^2$ especially transparent:

- in F_2 , the generators a, b do not satisfy a commutativity relation;
- in $\mathbb{Z}^2$, one has the relation $ab = ba$.

10.15 Covering spaces and the fundamental group

The fundamental group is closely related to covering spaces.

Examples.

- The universal cover of S^1 is $\mathbb{R}$.
- The universal cover of $S^1 \vee S^1$ is an infinite tree.
- Subgroups of $\pi_1(X)$ correspond, under suitable hypotheses, to connected covering spaces of X .

This viewpoint gives a topological proof of the Nielsen–Schreier theorem: every subgroup of a free group is free.

10.16 Why the fundamental group matters

The fundamental group is useful because it can:

- distinguish spaces that are not homotopy equivalent;
- detect holes and nontrivial winding;
- translate geometric questions into algebraic ones;
- connect topology with group theory through covering spaces, free groups, and presentations.

For example:

- S^1 and S^2 are not homotopy equivalent, because

$$\pi_1(S^1) \cong \mathbb{Z}, \quad \pi_1(S^2) \cong \{e\}.$$

- The unlink and the Hopf link are not equivalent in S^3 , because

$$\pi_1(S^3 \setminus \text{unlink}) \cong F_2, \quad \pi_1(S^3 \setminus \text{Hopf link}) \cong \mathbb{Z}^2.$$

- Every connected finite graph has a free fundamental group whose rank is determined by its combinatorics.

10.17 Summary

The fundamental group $\pi_1(X, x_0)$ is the group of based loops in X up to homotopy. It is one of the first and most powerful topological invariants.

Its basic examples already reveal much of topology:

$$\pi_1(\mathbb{R}^n) = 0, \quad \pi_1(S^1) \cong \mathbb{Z}, \quad \pi_1(T^2) \cong \mathbb{Z}^2, \quad \pi_1\left(\bigvee^n S^1\right) \cong F_n.$$

In the study of graphs, link complements, and covering spaces, the fundamental group serves as a bridge between geometry and algebra.

11 Problem 1 Abstract simplicial complexes, geometric realizations, and realization in Euclidean space

11.1 Definitions and background

Abstract simplicial complex. An *abstract simplicial complex* consists of a finite set V_X (called the *vertices*) and a collection X of subsets of V_X (called the *simplices*) such that

$$\sigma \in X, \tau \subseteq \sigma \implies \tau \in X.$$

Thus X is closed under taking subsets.

If $\sigma \in X$ has $k + 1$ vertices, then σ is said to have *dimension* k .

Map of abstract simplicial complexes. A map

$$f : (V_X, X) \rightarrow (V_Y, Y)$$

of abstract simplicial complexes is a function

$$f : V_X \rightarrow V_Y$$

such that for every simplex $\sigma \in X$, the image set $f(\sigma)$ is a simplex of Y .

Geometric simplicial complex. A *simplicial complex* K in $\mathbb{R}^m$ is a collection of geometric simplices in $\mathbb{R}^m$ such that:

- every face of a simplex of K again lies in K ;
- the intersection of any two simplices of K is either empty or a common face of both.

Its underlying topological space is denoted by

$$|K| = \bigcup_{\sigma \in K} \sigma.$$

Abstraction of a geometric simplicial complex. If K is a geometric simplicial complex, its *abstraction* is the abstract simplicial complex (V_X, X) where

$$V_X = \{0\text{-simplices of } K\},$$

and

$$X = \{\{a_0, \dots, a_n\} \subseteq V_X \mid \langle a_0, \dots, a_n \rangle \in K\}.$$

11.2 Problem statement

Problem. An abstract simplicial complex consists of a finite set V_X (called the vertices) and a collection X (called the simplices) of subsets of V_X such that if $\sigma \in X$ and $\tau \subseteq \sigma$, then $\tau \in X$. A map

$$f : (V_X, X) \rightarrow (V_Y, Y)$$

of abstract simplicial complexes is a function

$$f : V_X \rightarrow V_Y$$

such that $f(\sigma) \in Y$ for all $\sigma \in X$.

- (i) For a simplicial complex K in $\mathbb{R}^m$, show that the abstraction of K ,

$$V_X = \{0\text{-simplices of } K\}, \quad X = \{\{a_0, \dots, a_n\} \subseteq V_X \mid \langle a_0, \dots, a_n \rangle \in K\},$$

is an abstract simplicial complex. Show that any abstract simplicial complex arises in this way (up to isomorphism), and that if simplicial complexes K and L have isomorphic abstractions, then $|K|$ and $|L|$ are homeomorphic.

- (ii) Show that there exists an infinite sequence of points $(x_1, x_2, \dots) \in \mathbb{R}^m$ such that any $(m + 1)$ of them are affinely independent. Hence show that if (V_X, X) is an abstract simplicial complex with all simplices of dimension $\leq n$, then there is a simplicial complex K in $\mathbb{R}^{2n+1}$ with abstraction isomorphic to (V_X, X) .

11.3 Solution

Part (i): the abstraction of a geometric simplicial complex.

Let K be a simplicial complex in $\mathbb{R}^m$. Define

$$V_X = \{\text{vertices of } K\},$$

and

$$X = \left\{ \{a_0, \dots, a_n\} \subseteq V_X \mid \langle a_0, \dots, a_n \rangle \in K \right\}.$$

We must show that (V_X, X) is an abstract simplicial complex.

Take any simplex

$$\sigma = \{a_0, \dots, a_n\} \in X.$$

By definition, the geometric simplex

$$\langle a_0, \dots, a_n \rangle$$

belongs to K . Now let $\tau \subseteq \sigma$. Then τ is obtained by choosing some subset of the vertices $a_0, \dots, a_n$, so the geometric simplex spanned by τ is a face of $\langle a_0, \dots, a_n \rangle$. Since every face of a simplex in K again belongs to K , it follows that

$$\langle \tau \rangle \in K.$$

Hence $\tau \in X$. Therefore X is closed under taking subsets, so (V_X, X) is indeed an abstract simplicial complex.

Every abstract simplicial complex arises this way.

Now let (V_X, X) be any abstract simplicial complex, where

$$V_X = \{v_1, \dots, v_r\}.$$

We construct a geometric simplicial complex whose abstraction is isomorphic to (V_X, X) .

Place the vertices at the standard basis vectors

$$e_1, \dots, e_r \in \mathbb{R}^r.$$

For each simplex

$$\sigma = \{v_{i_0}, \dots, v_{i_k}\} \in X,$$

consider the geometric simplex

$$\langle e_{i_0}, \dots, e_{i_k} \rangle.$$

Let K be the collection of all such simplices.

We claim that K is a simplicial complex.

First, if

$$\langle e_{i_0}, \dots, e_{i_k} \rangle \in K,$$

then any face is of the form

$$\langle e_{j_0}, \dots, e_{j_\ell} \rangle$$

for some subset $\{v_{j_0}, \dots, v_{j_\ell}\} \subseteq \{v_{i_0}, \dots, v_{i_k}\}$. Since X is an abstract simplicial complex, every subset of a simplex is again a simplex, so

$$\{v_{j_0}, \dots, v_{j_\ell}\} \in X.$$

Hence this face also belongs to K .

Second, let

$$\sigma = \{v_{i_0}, \dots, v_{i_k}\}, \quad \tau = \{v_{j_0}, \dots, v_{j_\ell}\}$$

be simplices of X . Then

$$\langle e_{i_0}, \dots, e_{i_k} \rangle \cap \langle e_{j_0}, \dots, e_{j_\ell} \rangle = \langle e_s : v_s \in \sigma \cap \tau \rangle,$$

which is a face of both simplices. Thus K satisfies the intersection condition for a simplicial complex.

So K is a simplicial complex in $\mathbb{R}^r$, and by construction its abstraction is exactly (V_X, X) , up to the obvious identification $v_i \leftrightarrow e_i$.

Therefore every abstract simplicial complex arises as the abstraction of a geometric simplicial complex, up to isomorphism.

Isomorphic abstractions imply homeomorphic realizations.

Suppose simplicial complexes K and L have isomorphic abstractions. Then there is a bijection

$$\phi : V(K) \rightarrow V(L)$$

such that a finite set of vertices spans a simplex of K if and only if its image under ϕ spans a simplex of L .

We define a map

$$|\phi| : |K| \rightarrow |L|$$

simplexwise by barycentric coordinates. If

$$x = \sum_{i=0}^k t_i a_i \in \langle a_0, \dots, a_k \rangle \subseteq |K|, \quad t_i \geq 0, \quad \sum_{i=0}^k t_i = 1,$$

then set

$$|\phi|(x) = \sum_{i=0}^k t_i \phi(a_i) \in \langle \phi(a_0), \dots, \phi(a_k) \rangle \subseteq |L|.$$

This is well-defined, because ϕ sends simplices of K to simplices of L , and on intersections of simplices the barycentric coordinate descriptions agree. On each simplex, $|\phi|$ is an affine homeomorphism. Likewise, ϕ^{-1} induces a map

$$|\phi^{-1}| : |L| \rightarrow |K|$$

which is inverse to $|\phi|$.

Hence $|\phi| : |K| \rightarrow |L|$ is a homeomorphism.

Therefore, if two simplicial complexes have isomorphic abstractions, then their realizations are homeomorphic:

$$|K| \cong |L|.$$

Part (ii): infinite affine-independent sequences and realization in $\mathbb{R}^{2n+1}$.

We first prove the existence of an infinite sequence of points in $\mathbb{R}^m$ such that any $m + 1$ of them are affinely independent.

Consider the *moment curve*

$$\gamma(t) = (t, t^2, \dots, t^m) \in \mathbb{R}^m.$$

Choose distinct real numbers

$$t_1, t_2, t_3, \dots$$

and define

$$x_i = \gamma(t_i) = (t_i, t_i^2, \dots, t_i^m) \in \mathbb{R}^m.$$

We claim that any $m + 1$ of these points are affinely independent.

Indeed, points $x_{i_0}, \dots, x_{i_m} \in \mathbb{R}^m$ are affinely independent if and only if the vectors

$$(x_{i_0}, 1), \dots, (x_{i_m}, 1) \in \mathbb{R}^{m+1}$$

are linearly independent. But these vectors form the rows of the matrix

$$\begin{pmatrix} 1 & t_{i_0} & t_{i_0}^2 & \dots & t_{i_0}^m \\ 1 & t_{i_1} & t_{i_1}^2 & \dots & t_{i_1}^m \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & t_{i_m} & t_{i_m}^2 & \dots & t_{i_m}^m \end{pmatrix}.$$

This is a Vandermonde matrix, whose determinant is

$$\prod_{p < q} (t_{i_q} - t_{i_p}).$$

Since the t_{i_j} are distinct, this determinant is nonzero. Therefore the vectors are linearly independent, and the points $x_{i_0}, \dots, x_{i_m}$ are affinely independent.

Thus there exists an infinite sequence of points in $\mathbb{R}^m$ such that any $m + 1$ of them are affinely independent.

Realization in $\mathbb{R}^{2n+1}$.

Now let (V_X, X) be an abstract simplicial complex such that every simplex has dimension at most n . Equivalently, every simplex has at most $n + 1$ vertices.

Write

$$V_X = \{v_1, \dots, v_r\}.$$

Choose points

$$x_1, \dots, x_r \in \mathbb{R}^{2n+1}$$

such that any $2n + 2$ of them are affinely independent. This is possible by the first part, applied with $m = 2n + 1$, for example by choosing the x_i on the moment curve in $\mathbb{R}^{2n+1}$.

For each simplex

$$\sigma = \{v_{i_0}, \dots, v_{i_k}\} \in X, \quad k \leq n,$$

let

$$\langle x_{i_0}, \dots, x_{i_k} \rangle$$

be the geometric simplex spanned by the corresponding points. Let K be the collection of all such simplices.

We show that K is a simplicial complex in $\mathbb{R}^{2n+1}$.

First, every face of a simplex of K again belongs to K , because if

$$\{v_{i_0}, \dots, v_{i_k}\} \in X,$$

then every subset is again in X , since X is an abstract simplicial complex.

So only the intersection condition remains.

Lemma. If A and B are finite sets of points in $\mathbb{R}^N$, and $A \cup B$ is affinely independent, then

$$\text{conv}(A) \cap \text{conv}(B) = \text{conv}(A \cap B).$$

Proof. Certainly

$$\text{conv}(A \cap B) \subseteq \text{conv}(A) \cap \text{conv}(B).$$

For the reverse inclusion, let

$$x \in \text{conv}(A) \cap \text{conv}(B).$$

Then

$$x = \sum_{a \in A} \lambda_a a = \sum_{b \in B} \mu_b b, \quad \lambda_a, \mu_b \geq 0, \quad \sum_{a \in A} \lambda_a = \sum_{b \in B} \mu_b = 1.$$

Subtracting gives

$$\sum_{a \in A \setminus B} \lambda_a a + \sum_{c \in A \cap B} (\lambda_c - \mu_c) c - \sum_{b \in B \setminus A} \mu_b b = 0.$$

Also the sum of the coefficients here is

$$\sum_{a \in A \setminus B} \lambda_a + \sum_{c \in A \cap B} (\lambda_c - \mu_c) - \sum_{b \in B \setminus A} \mu_b = 1 - 1 = 0.$$

So this is an affine dependence relation among the points of $A \cup B$. Since $A \cup B$ is affinely independent, all coefficients must be zero. Hence

$$\lambda_a = 0 \text{ for } a \in A \setminus B, \quad \mu_b = 0 \text{ for } b \in B \setminus A.$$

Therefore

$$x \in \text{conv}(A \cap B).$$

So indeed

$$\text{conv}(A) \cap \text{conv}(B) = \text{conv}(A \cap B).$$

This proves the lemma.

Now apply the lemma to two simplices of K , say corresponding to

$$\sigma, \tau \in X.$$

Since each simplex has at most $n + 1$ vertices,

$$|\sigma \cup \tau| \leq (n + 1) + (n + 1) = 2n + 2.$$

By construction, any $2n + 2$ of the chosen points are affinely independent, so the set of vertices corresponding to $\sigma \cup \tau$ is affinely independent. Therefore the lemma gives

$$\langle \sigma \rangle \cap \langle \tau \rangle = \langle \sigma \cap \tau \rangle,$$

that is, the intersection of any two simplices of K is a common face.

Hence K is a simplicial complex in $\mathbb{R}^{2n+1}$.

Finally, by construction, the abstraction of K is isomorphic to (V_X, X) : the vertices of K are in bijection with the vertices $v_1, \dots, v_r$, and a subset of vertices spans a simplex of K exactly when the corresponding subset belongs to X .

Thus there exists a simplicial complex

$$K \subseteq \mathbb{R}^{2n+1}$$

whose abstraction is isomorphic to (V_X, X) .

11.4 Conclusion

We have proved the following:

- The abstraction of a geometric simplicial complex is an abstract simplicial complex.
- Every abstract simplicial complex can be realized geometrically, up to isomorphism.
- If two simplicial complexes have isomorphic abstractions, then their realizations are homeomorphic.
- If all simplices have dimension at most n , then the abstract simplicial complex can be realized by a simplicial complex in $\mathbb{R}^{2n+1}$.

So abstract simplicial complexes and geometric simplicial complexes encode the same combinatorial data, and bounded dimension allows realization in a controlled Euclidean dimension.

12 Problem 2 Triangulations of the torus, Klein bottle, and projective plane

12.1 Problem statement

Problem. Show that there are triangulations of the torus, Klein bottle, and projective plane as indicated in the figure.

How many vertices, edges, and faces does each triangulation have? What is the number

$$\chi = \text{vertices} - \text{edges} + \text{faces}$$

for each triangulation?

12.2 Solution

We count the numbers of vertices, edges, and triangular faces in each quotient triangulation.

1. The torus and Klein bottle

In the figure for the torus and Klein bottle, the square is divided into a 3×3 grid, and each small square is cut by a diagonal. Thus before taking the quotient, the square is triangulated as follows:

- vertices: $4 \times 4 = 16$,
- small squares: $3 \times 3 = 9$,
- triangles: $2 \cdot 9 = 18$.

So the number of faces after quotienting is still

$$F = 18,$$

since no two triangles are identified with one another.

Counting vertices.

There are:

- 4 corner vertices,
- 8 non-corner boundary vertices,
- 4 interior vertices.

For both the torus and the Klein bottle, the four corners are all identified to a single vertex in the quotient.

The 8 non-corner boundary vertices are identified in pairs under the side gluings, giving 4 boundary vertex classes.

The 4 interior vertices remain distinct.

Hence the number of vertices is

$$V = 1 + 4 + 4 = 9.$$

Counting edges.

Before quotienting, the subdivided square has:

- 12 horizontal edges,
- 12 vertical edges,
- 9 diagonal edges.

Thus in total there are

$$12 + 12 + 9 = 33$$

edges in the square triangulation.

Now pass to the quotient.

- Among the horizontal edges, 6 lie on the top and bottom boundary and are identified in pairs, so they give 3 quotient edges. The remaining 6 horizontal edges are interior and remain distinct.
- Among the vertical edges, 6 lie on the left and right boundary and are identified in pairs, so they give 3 quotient edges. The remaining 6 vertical edges are interior and remain distinct.
- All 9 diagonal edges remain distinct.

Therefore

$$E = 3 + 6 + 3 + 6 + 9 = 27.$$

So for both the torus and the Klein bottle we obtain

$$V = 9, \quad E = 27, \quad F = 18.$$

Hence

$$\chi = V - E + F = 9 - 27 + 18 = 0.$$

Thus:

$$\boxed{\chi(T) = 0}, \quad \boxed{\chi(K) = 0}.$$

2. The projective plane

Now consider the triangulation shown for $\mathbb{RP}^2$.

We first count the triangles in the square. From the drawing, the square is divided into 10 triangles, so

$$F = 10.$$

Counting vertices before quotient.

In the square picture there are 9 vertices:

- the 4 corner vertices,
- 2 additional boundary vertices: one on the left side and one on the right side,
- 3 interior vertices.

Under the projective-plane identifications:

- the four corners split into two equivalence classes:

$$\{\text{top-left, bottom-right}\}, \quad \{\text{top-right, bottom-left}\},$$

- the two extra boundary vertices are identified with each other,
- the 3 interior vertices remain distinct.

Therefore the number of vertices in the quotient is

$$V = 2 + 1 + 3 = 6.$$

Counting edges before quotient.

In the square triangulation there are 18 edges altogether. This can be checked directly, or by using Euler's formula for the disk:

$$V - E + F = 1.$$

Since $V = 9$ and $F = 10$, we get

$$9 - E + 10 = 1,$$

so

$$E = 18.$$

Among these 18 edges, 6 lie on the boundary of the square, and the remaining 12 are interior.

Under the projective-plane identifications, the 6 boundary edges are identified in pairs, so they become 3 quotient edges. The 12 interior edges remain distinct.

Hence

$$E = 12 + 3 = 15.$$

Thus for the projective plane:

$$V = 6, \quad E = 15, \quad F = 10.$$

Therefore

$$\chi = V - E + F = 6 - 15 + 10 = 1.$$

So

$$\boxed{\chi(\mathbb{RP}^2) = 1}.$$

12.3 Final counts

Surface	V	E	F	$\chi = V - E + F$
T	9	27	18	0
K	9	27	18	0
$\mathbb{RP}^2$	6	15	10	1

12.4 Conclusion

For the triangulations shown in the figure, we have:

$$\boxed{\begin{array}{ll} T : (V, E, F) = (9, 27, 18), & \chi(T) = 0, \\ K : (V, E, F) = (9, 27, 18), & \chi(K) = 0, \\ \mathbb{RP}^2 : (V, E, F) = (6, 15, 10), & \chi(\mathbb{RP}^2) = 1. \end{array}}$$

These values agree with the known Euler characteristics of the torus, Klein bottle, and projective plane.

13 Problem 3 Simplicial approximation: countability of homotopy classes, maps between spheres, and the 2-skeleton

13.1 Definitions and background

Simplicial approximation theorem. Let K and L be simplicial complexes, and let

$$f : |K| \rightarrow |L|$$

be continuous. Then after sufficiently many barycentric subdivisions of K , there exists a simplicial map

$$g : \text{sd}^r K \rightarrow L$$

such that $|g|$ is homotopic to f .

Thus, up to homotopy, continuous maps between polyhedra may be replaced by simplicial maps from sufficiently fine subdivisions.

Skeletons. If K is a simplicial complex, its n -skeleton $K^{(n)}$ is the subcomplex consisting of all simplices of dimension at most n .

Idea. The simplicial approximation theorem lets one reduce topological questions about arbitrary continuous maps to combinatorial questions about simplicial maps. This is especially effective when the domain has small dimension, since a simplicial map from a low-dimensional complex can only land in a low-dimensional skeleton of the target.

13.2 Problem statement

Problem. Use the simplicial approximation theorem to show that

- (i) if K and L are simplicial complexes, there are at most countably many homotopy classes of continuous maps

$$f : |K| \rightarrow |L|;$$

- (ii) if $m < n$, then any continuous map

$$S^m \rightarrow S^n$$

is homotopic to a constant map;

- (iii) for any vertex v of a simplicial complex K , the based map

$$(|K^{(2)}|, v) \rightarrow (|K|, v)$$

(i.e. the inclusion of the 2-skeleton) induces an isomorphism on fundamental groups. Show how to prove the same result using the Seifert–van Kampen theorem.

13.3 Solution

(i) **There are at most countably many homotopy classes of maps $|K| \rightarrow |L|$.**

Let

$$f : |K| \rightarrow |L|$$

be continuous. By the simplicial approximation theorem, there exists some integer $r \geq 0$ such that after r barycentric subdivisions of K , the map f is homotopic to the realization of a simplicial map

$$g : \text{sd}^r K \rightarrow L.$$

Hence every homotopy class of continuous maps contains a simplicial representative from some subdivision $\text{sd}^r K$.

Now fix r . A simplicial map

$$g : \text{sd}^r K \rightarrow L$$

is completely determined by its effect on vertices, subject to the simplicial condition that the image of each simplex is again a simplex. If K and L are finite, then $\text{sd}^r K$ and L have only finitely many vertices, so for fixed r there are only finitely many such simplicial maps. More generally, if the complexes are countable, then for fixed r there are at most countably many simplicial maps.

Therefore, for each fixed r , the set of simplicial maps

$$\text{sd}^r K \rightarrow L$$

is at most countable. Since r ranges over the countable set $\{0, 1, 2, \dots\}$, the total set of simplicial maps from all subdivisions

$$\bigcup_{r \geq 0} \text{Map}_{\text{simp}}(\text{sd}^r K, L)$$

is at most countable.

Since every homotopy class of continuous maps $|K| \rightarrow |L|$ contains one of these simplicial representatives, it follows that there are at most countably many homotopy classes of continuous maps

$$|K| \rightarrow |L|.$$

Conclusion for (i). The set

$$[|K|, |L|]$$

of homotopy classes of continuous maps is at most countable.

(ii) **If $m < n$, every map $S^m \rightarrow S^n$ is null-homotopic.**

Let

$$f : S^m \rightarrow S^n$$

be continuous. Choose triangulations

$$|K| \cong S^m, \quad |L| \cong S^n.$$

By simplicial approximation, after sufficiently many barycentric subdivisions of K , the map f is homotopic to the realization of a simplicial map

$$g : \text{sd}^r K \rightarrow L.$$

Now $\text{sd}^r K$ is still an m -dimensional simplicial complex. Therefore the image of each simplex of $\text{sd}^r K$ under g is a simplex of L of dimension at most m . Hence the image of g is contained in the m -skeleton

$$L^{(m)} \subseteq L.$$

Since $m < n$, the m -skeleton $L^{(m)}$ is a proper subcomplex of L , so it cannot fill all of $|L| = S^n$. Therefore there exists some point

$$y \in S^n$$

which does not belong to the image of g . Thus

$$g : S^m \rightarrow S^n \setminus \{y\}.$$

But

$$S^n \setminus \{y\} \cong \mathbb{R}^n,$$

and $\mathbb{R}^n$ is contractible. Therefore g is homotopic to a constant map. Since $f \simeq g$, it follows that f is also homotopic to a constant map.

Conclusion for (ii). If $m < n$, every continuous map

$$S^m \rightarrow S^n$$

is null-homotopic.

(iii) The inclusion $|K^{(2)}| \hookrightarrow |K|$ induces an isomorphism on fundamental groups.

Let

$$i : (|K^{(2)}|, v) \hookrightarrow (|K|, v)$$

be the inclusion.

We show that

$$i_* : \pi_1(|K^{(2)}|, v) \rightarrow \pi_1(|K|, v)$$

is an isomorphism.

Surjectivity. Let

$$\alpha : (S^1, *) \rightarrow (|K|, v)$$

be a based loop. Triangulate S^1 as a 1-dimensional simplicial complex, with the basepoint as a vertex. By simplicial approximation, after subdividing S^1 sufficiently many times, the loop α is homotopic through based maps to the realization of a simplicial map

$$\beta : \text{sd}^r(S^1) \rightarrow K.$$

Since the domain is 1-dimensional, the image of β lies in the 1-skeleton $K^{(1)}$, hence certainly in $K^{(2)}$. Thus every loop in $|K|$ is homotopic to a loop in $|K^{(2)}|$. Therefore

$$i_* : \pi_1(|K^{(2)}|, v) \rightarrow \pi_1(|K|, v)$$

is surjective.

Injectivity. Now suppose two based loops in $|K^{(2)}|$ are homotopic in $|K|$. Then there exists a based homotopy

$$H : S^1 \times I \rightarrow |K|$$

between them. Triangulate the cylinder $S^1 \times I$ as a 2-dimensional simplicial complex, with boundary a subcomplex. By the relative form of simplicial approximation, after subdividing sufficiently many times, the homotopy H is homotopic relative to the boundary to the realization of a simplicial map

$$H' : \text{sd}^r(S^1 \times I) \rightarrow K$$

which agrees with the given loops on the boundary.

Since the domain is 2-dimensional, the image of H' lies in the 2-skeleton $K^{(2)}$. Thus the original two loops are homotopic already inside $|K^{(2)}|$. Therefore

$$i_* : \pi_1(|K^{(2)}|, v) \rightarrow \pi_1(|K|, v)$$

is injective.

Combining injectivity and surjectivity, we conclude that

$$i_* : \pi_1(|K^{(2)}|, v) \xrightarrow{\cong} \pi_1(|K|, v)$$

is an isomorphism.

The same result via the Seifert–van Kampen theorem.

There is also a more geometric proof. Start with the 2-skeleton $K^{(2)}$, and obtain K by attaching simplices of dimensions 3, 4, ...

Suppose τ is a simplex of dimension $q \geq 3$. Topologically, τ is a q -disk D^q , and it is attached along its boundary

$$\partial\tau \cong S^{q-1}.$$

Since $q - 1 \geq 2$, the sphere S^{q-1} is simply connected. Also D^q is contractible, hence simply connected. When one attaches τ to the space already constructed, the Seifert–van Kampen theorem shows that the fundamental group does not change, because one is gluing along a path-connected, simply connected subspace.

More explicitly, if

$$X = Y \cup D^q$$

with

$$Y \cap D^q = S^{q-1},$$

then van Kampen gives

$$\pi_1(X) \cong \pi_1(Y) *_{\pi_1(S^{q-1})} \pi_1(D^q).$$

But

$$\pi_1(S^{q-1}) = 0 \quad \text{and} \quad \pi_1(D^q) = 0,$$

so

$$\pi_1(X) \cong \pi_1(Y).$$

Thus attaching simplices of dimension at least 3 does not change the fundamental group.

Applying this repeatedly, we conclude that passing from $K^{(2)}$ to K leaves the fundamental group unchanged:

$$\pi_1(|K^{(2)}|, v) \cong \pi_1(|K|, v).$$

13.4 Conclusion

We have proved:

- there are at most countably many homotopy classes of maps $|K| \rightarrow |L|$;
- if $m < n$, every map $S^m \rightarrow S^n$ is null-homotopic;
- the inclusion

$$|K^{(2)}| \hookrightarrow |K|$$

induces an isomorphism on fundamental groups.

These are typical and important consequences of the simplicial approximation theorem: low-dimensional homotopy information is controlled by low-dimensional skeleta, and continuous maps may be studied combinatorially after subdivision.

14 Problem 4 Deleting a maximal simplex and the resulting homology exact sequence

14.1 Definitions and background

Deleting a simplex. Let K be a simplicial complex, and let $\sigma \in K$. If σ is not a proper face of any simplex of K , then σ is maximal with respect to inclusion. Removing such a simplex from K changes the complex in a very controlled way.

Relative homology viewpoint. If

$$L = K \setminus \{\sigma\},$$

then the pair (K, L) differs only by one simplex. Therefore the relative chain complex $C_*(K, L)$ is concentrated in one degree, and the long exact sequence of the pair (K, L) describes precisely how the homology changes when σ is removed.

14.2 Problem statement

Problem. Let K be a simplicial complex, and suppose that $\sigma \in K$ is not a proper face of any simplex. Show that

$$L = K \setminus \{\sigma\}$$

is again a simplicial complex, and that the inclusion $V_L \rightarrow V_K$ defines a simplicial map

$$i : L \rightarrow K.$$

If σ has dimension n , note that $d_n(\sigma)$ is an $(n-1)$ -cycle and consists of simplices of L , so represents a class

$$[d_n(\sigma)] \in H_{n-1}(L);$$

this defines a homomorphism

$$\varphi : \mathbb{Z} \rightarrow H_{n-1}(L)$$

by

$$1 \mapsto [d_n(\sigma)].$$

Construct a homomorphism

$$\phi : H_n(K) \rightarrow \mathbb{Z}$$

such that

$$0 \longrightarrow H_n(L) \xrightarrow{i_*} H_n(K) \xrightarrow{\phi} \mathbb{Z} \xrightarrow{\varphi} H_{n-1}(L) \xrightarrow{i_*} H_{n-1}(K) \longrightarrow 0$$

is exact, and show that

$$i_* : H_j(L) \rightarrow H_j(K)$$

is an isomorphism for $j \neq n-1, n$.

14.3 Solution

Step 1: $L = K \setminus \{\sigma\}$ is again a simplicial complex.

We must show that L is closed under taking faces.

Let $\tau \in L$. Then $\tau \in K$ and $\tau \neq \sigma$. Let $\eta \subseteq \tau$. Since K is a simplicial complex, we already know that $\eta \in K$. We must show that $\eta \neq \sigma$, so that $\eta \in L$.

If $\eta = \sigma$, then

$$\sigma \subseteq \tau.$$

Since $\tau \neq \sigma$, this would mean that σ is a proper face of the simplex τ , contradicting the hypothesis that σ is not a proper face of any simplex of K .

Therefore $\eta \neq \sigma$, so $\eta \in L$. Hence L is a simplicial complex.

Step 2: the inclusion $V_L \rightarrow V_K$ defines a simplicial map $i : L \rightarrow K$.

The vertex set of L is contained in the vertex set of K , so there is a natural inclusion

$$V_L \hookrightarrow V_K.$$

If $\tau \in L$, then τ is by definition also a simplex of K . Therefore the image of every simplex of L is a simplex of K . So the inclusion of vertex sets defines a simplicial map

$$i : L \rightarrow K.$$

Step 3: the relative chain complex $C_*(K, L)$.

Let $\dim \sigma = n$.

Because σ is not a proper face of any simplex of K , the only simplex present in K and not in L is σ itself. Therefore the simplicial chain groups satisfy:

$$C_j(K) = C_j(L) \quad \text{for } j \neq n,$$

while in degree n ,

$$C_n(K) = C_n(L) \oplus \mathbb{Z}\langle\sigma\rangle.$$

Hence the relative chain groups are

$$C_j(K, L) \cong \begin{cases} \mathbb{Z}, & j = n, \\ 0, & j \neq n. \end{cases}$$

Moreover, since σ is not a face of any $(n+1)$ -simplex, the relative boundary map is zero. Thus

$$H_j(K, L) \cong \begin{cases} \mathbb{Z}, & j = n, \\ 0, & j \neq n. \end{cases}$$

So the pair (K, L) has relative homology concentrated in degree n , where it is generated by the class of the simplex σ .

Step 4: the homomorphism $\varphi : \mathbb{Z} \rightarrow H_{n-1}(L)$.

The simplicial boundary

$$d_n(\sigma) = \partial\sigma$$

is a sum of the $(n-1)$ -faces of σ . Since only the simplex σ itself was removed, all of its proper faces remain in L . Therefore $d_n(\sigma)$ is a simplicial $(n-1)$ -chain in L .

Since

$$d_{n-1}d_n(\sigma) = 0,$$

the chain $d_n(\sigma)$ is an $(n-1)$ -cycle in L , and so it defines a homology class

$$[d_n(\sigma)] \in H_{n-1}(L).$$

This gives a homomorphism

$$\varphi : \mathbb{Z} \rightarrow H_{n-1}(L), \quad 1 \mapsto [d_n(\sigma)].$$

Under the identification

$$H_n(K, L) \cong \mathbb{Z}$$

with generator represented by the class of σ , this map φ is exactly the connecting homomorphism in the long exact sequence of the pair (K, L) .

Step 5: construction of $\phi : H_n(K) \rightarrow \mathbb{Z}$.

The long exact sequence of the pair (K, L) contains the map

$$H_n(K) \rightarrow H_n(K, L).$$

Since

$$H_n(K, L) \cong \mathbb{Z},$$

we define

$$\phi : H_n(K) \rightarrow \mathbb{Z}$$

to be this map, followed by the chosen identification of $H_n(K, L)$ with $\mathbb{Z}$.

Concretely, if a class $[z] \in H_n(K)$ is represented by an n -cycle of the form

$$z = a\sigma + c, \quad c \in C_n(L),$$

then

$$\phi([z]) = a.$$

In other words, ϕ records the coefficient of the deleted simplex σ .

This is well-defined because no $(n+1)$ -simplex has σ as a face, so boundaries in K cannot change the coefficient of σ .

Step 6: the exact sequence.

Now apply the long exact sequence of the pair (K, L) :

$$\cdots \rightarrow H_n(L) \xrightarrow{i_*} H_n(K) \xrightarrow{\phi} H_n(K, L) \xrightarrow{\partial} H_{n-1}(L) \xrightarrow{i_*} H_{n-1}(K) \rightarrow \cdots$$

Using the identification

$$H_n(K, L) \cong \mathbb{Z}$$

and the fact that

$$H_j(K, L) = 0 \quad \text{for } j \neq n,$$

this becomes

$$0 \longrightarrow H_n(L) \xrightarrow{i_*} H_n(K) \xrightarrow{\phi} \mathbb{Z} \xrightarrow{\varphi} H_{n-1}(L) \xrightarrow{i_*} H_{n-1}(K) \longrightarrow 0.$$

This is the desired exact sequence.

Interpretation of exactness.

At $H_n(K)$, exactness says that

$$\ker \phi = \text{im}(i_*).$$

So an n -homology class in K comes from L exactly when its representative has zero coefficient on σ .

At $\mathbb{Z}$, exactness says

$$\ker \varphi = \text{im}(\phi).$$

Thus an integer a lies in the image of ϕ exactly when the cycle

$$a d_n(\sigma)$$

is homologous to zero in L .

At $H_{n-1}(L)$, exactness says

$$\ker(i_*) = \text{im}(\varphi).$$

So the only $(n-1)$ -homology classes in L that vanish in K are those generated by the boundary of the deleted simplex.

Step 7: isomorphisms in all other degrees.

Because

$$H_j(K, L) = 0 \quad \text{for } j \neq n,$$

the long exact sequence of the pair shows that for every $j \neq n, n-1$, the map

$$i_* : H_j(L) \rightarrow H_j(K)$$

is an isomorphism. Indeed, the relevant segment of the long exact sequence takes the form

$$0 \rightarrow H_j(L) \xrightarrow{i_*} H_j(K) \rightarrow 0.$$

Therefore,

$$i_* : H_j(L) \xrightarrow{\cong} H_j(K) \quad \text{for all } j \neq n-1, n.$$

14.4 Conclusion

We have proved the following:

- If $\sigma \in K$ is not a proper face of any simplex, then

$$L = K \setminus \{\sigma\}$$

is again a simplicial complex.

- The inclusion of vertex sets defines a simplicial map

$$i : L \rightarrow K.$$

- If $\dim \sigma = n$, then there is an exact sequence

$$0 \longrightarrow H_n(L) \xrightarrow{i_*} H_n(K) \xrightarrow{\phi} \mathbb{Z} \xrightarrow{\varphi} H_{n-1}(L) \xrightarrow{i_*} H_{n-1}(K) \longrightarrow 0,$$

where

$$\varphi(1) = [d_n(\sigma)].$$

- For every degree $j \neq n-1, n$, the map

$$i_* : H_j(L) \rightarrow H_j(K)$$

is an isomorphism.

So removing a maximal simplex affects homology only in the two adjacent degrees n and $n-1$, and the effect is described completely by the boundary of the removed simplex.

14.5 Examples for deleting a maximal simplex

We now give several concrete examples illustrating Problem 4.

Recall the setup: K is a simplicial complex, $\sigma \in K$ is a simplex which is not a proper face of any simplex, and

$$L = K \setminus \{\sigma\}.$$

If $\dim \sigma = n$, then there is an exact sequence

$$0 \longrightarrow H_n(L) \xrightarrow{i_*} H_n(K) \xrightarrow{\phi} \mathbb{Z} \xrightarrow{\varphi} H_{n-1}(L) \xrightarrow{i_*} H_{n-1}(K) \longrightarrow 0,$$

and

$$i_* : H_j(L) \rightarrow H_j(K)$$

is an isomorphism for all $j \neq n, n-1$.

Thus removing a maximal n -simplex can affect homology only in degrees n and $n-1$.

Example 1: removing the only edge from an interval.

Let K be the simplicial complex consisting of one edge $[v_0v_1]$ together with its two vertices. Take

$$\sigma = [v_0v_1].$$

Then σ is maximal, and

$$L = K \setminus \{[v_0v_1]\}$$

consists of the two vertices $[v_0]$ and $[v_1]$, with no edge between them.

Topologically,

$$|K| \cong [0, 1], \quad |L| = \{v_0, v_1\}.$$

Hence

$$H_1(K) = 0, \quad H_1(L) = 0,$$

while

$$H_0(K) \cong \mathbb{Z}, \quad H_0(L) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

The exact sequence becomes

$$0 \rightarrow 0 \rightarrow 0 \rightarrow \mathbb{Z} \xrightarrow{\varphi} \mathbb{Z} \oplus \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0.$$

Here

$$\partial[v_0v_1] = [v_1] - [v_0],$$

so

$$\varphi(1) = [v_1] - [v_0].$$

This example shows that removing a maximal 1-simplex may increase the number of connected components, hence change H_0 .

Example 2: removing the filled triangle from a 2-simplex.

Let K be the full 2-simplex $[abc]$, together with all of its faces. Take

$$\sigma = [abc].$$

Then

$$L = K \setminus \{[abc]\}$$

is the boundary of the triangle:

$$L = \partial[abc] \cong S^1.$$

Thus

$$H_2(K) = 0, \quad H_2(L) = 0,$$

and

$$H_1(K) = 0, \quad H_1(L) \cong \mathbb{Z}.$$

The exact sequence becomes

$$0 \rightarrow 0 \rightarrow 0 \rightarrow \mathbb{Z} \xrightarrow{\varphi} \mathbb{Z} \rightarrow 0 \rightarrow 0,$$

so φ is an isomorphism. Indeed,

$$\partial[abc] = [bc] - [ac] + [ab],$$

and this is the fundamental 1-cycle of the circle $\partial[abc]$.

This example shows that removing a maximal 2-simplex may create a nontrivial 1-dimensional homology class.

Example 3: removing the filled tetrahedron from a 3-simplex.

Let K be the full 3-simplex $[abcd]$, together with all of its faces. Take

$$\sigma = [abcd].$$

Then

$$L = K \setminus \{[abcd]\}$$

is the boundary of the tetrahedron:

$$L = \partial[abcd] \cong S^2.$$

Hence

$$H_3(K) = 0, \quad H_3(L) = 0,$$

while

$$H_2(K) = 0, \quad H_2(L) \cong \mathbb{Z}.$$

The exact sequence becomes

$$0 \rightarrow 0 \rightarrow 0 \rightarrow \mathbb{Z} \xrightarrow{\varphi} \mathbb{Z} \rightarrow 0 \rightarrow 0.$$

Again φ is an isomorphism, and the boundary

$$\partial[abcd]$$

represents the fundamental class of the sphere $\partial[abcd]$.

This is the 3-dimensional analogue of the previous example: removing the top simplex exposes its boundary sphere and creates one lower-dimensional homology class.

Example 4: removing one face from a triangulated 2-sphere.

Let K be the boundary of a tetrahedron, so

$$|K| \cong S^2.$$

This complex has four triangular 2-simplices. Choose one of them, say

$$\sigma = [abc].$$

Since K is 2-dimensional, σ is not a proper face of any simplex.

Then

$$L = K \setminus \{\sigma\}$$

is the union of the remaining three faces, which is topologically a disk.

Thus

$$H_2(K) \cong \mathbb{Z}, \quad H_2(L) = 0,$$

and

$$H_1(K) = 0, \quad H_1(L) = 0.$$

The exact sequence becomes

$$0 \rightarrow 0 \rightarrow \mathbb{Z} \xrightarrow{\phi} \mathbb{Z} \rightarrow 0 \rightarrow 0.$$

So ϕ is an isomorphism.

This example shows the opposite behavior from the previous two: removing a maximal simplex may destroy a top-dimensional homology class rather than create a new one in one lower degree.

Example 5: removing a leaf edge from a graph with a loop.

Let K be a graph consisting of a circle with one extra dangling edge attached at a point. Let the dangling edge be

$$\sigma = [uv],$$

where v is the leaf vertex. Then

$$L = K \setminus \{\sigma\}$$

is the circle together with the isolated vertex v .

Topologically,

$$|K| \simeq S^1, \quad |L| \simeq S^1 \sqcup \{\text{point}\}.$$

Hence

$$H_1(K) \cong \mathbb{Z}, \quad H_1(L) \cong \mathbb{Z},$$

while

$$H_0(K) \cong \mathbb{Z}, \quad H_0(L) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

So in this example the higher homology remains unchanged, but the space becomes disconnected and H_0 changes.

Example 6: removing one triangle from a triangulated disk.

Let K be a disk triangulated by two triangles sharing an edge:

$$K = [abc] \cup [acd].$$

Take

$$\sigma = [abc].$$

Then

$$L = K \setminus \{\sigma\}$$

is the remaining triangle $[acd]$ together with all of its faces.

Both K and L are contractible. Hence

$$H_j(K) \cong H_j(L) \quad \text{for all } j.$$

In this case the exact sequence reads

$$0 \rightarrow H_2(L) \rightarrow H_2(K) \rightarrow \mathbb{Z} \xrightarrow{\varphi} H_1(L) \rightarrow H_1(K) \rightarrow 0,$$

but every homology group appearing here is zero. Thus removing σ does not change homology at all.

This example shows that deleting a maximal simplex need not change homology, even though Problem 4 allows changes in two adjacent degrees.

Example 7: general pattern seen in the examples.

The previous examples illustrate the main possibilities:

- Removing σ may create a new $(n - 1)$ -dimensional class, as in the cases

$$D^2 \setminus (\text{interior triangle}) \simeq S^1, \quad D^3 \setminus (\text{interior tetrahedron}) \simeq S^2.$$

- Removing σ may destroy a top-dimensional class, as when one removes one face from a triangulated sphere.
- Removing σ may leave all higher homology unchanged but alter connectivity, as in graph examples.
- Removing σ may leave all homology unchanged, as in a triangulated disk where one deletes one boundary triangle.

So the exact sequence of Problem 4 captures all possible changes caused by deleting a maximal simplex: only degrees n and $n - 1$ can be affected.

14.6 Summary table

For convenience, we summarize the above examples in a table:

K	σ	$L = K \setminus \{\sigma\}$	main homological effect
one edge	$[v_0v_1]$	$\{v_0, v_1\}$	H_0 increases
filled triangle	$[abc]$	$\partial[abc] \cong S^1$	H_1 appears
filled tetrahedron	$[abcd]$	$\partial[abcd] \cong S^2$	H_2 appears
S^2 triangulated	one face	D^2	H_2 disappears
circle + leaf edge	leaf edge	$S^1 \sqcup \{\text{pt}\}$	H_0 increases
two-triangle disk	one triangle	one triangle	no change

These examples provide a concrete guide to the exact sequence in Problem 4.

15 Problem 5 Elementary collapse of a simplex pair and invariance of homology

15.1 Definitions and background

Free face and elementary collapse. Let K be a simplicial complex. A simplex $\tau \in K$ is called a *free face* if there exists exactly one simplex $\sigma \in K$ such that

$$\tau < \sigma$$

and $\dim \sigma = \dim \tau + 1$, and τ is not a face of any other simplex of K . Deleting the pair (σ, τ) is called an *elementary collapse*.

Algebraic meaning. Geometrically, an elementary collapse removes a simplex together with a free codimension-one face. Topologically this does not change the homotopy type, and in particular it should not change homology. In this problem one proves that directly on simplicial chain complexes.

15.2 Problem statement

Problem. Let K be a simplicial complex, and suppose that $\sigma \in K$ is not a proper face of any simplex, and that $\tau \leq \sigma$ is a face of one dimension lower which is not a face of any other simplex. Show that

$$L = K \setminus \{\sigma, \tau\}$$

is again a simplicial complex, and that the inclusion

$$V_L \rightarrow V_K$$

defines a simplicial map

$$i : L \rightarrow K.$$

(i) By constructing a chain homotopy inverse to

$$i_{\#} : C_{\bullet}(L) \rightarrow C_{\bullet}(K),$$

show that

$$i_{*} : H_j(L) \rightarrow H_j(K)$$

is an isomorphism for all j .

(ii) Prove the same thing using the previous question twice instead.

15.3 Solution

Write

$$\dim \sigma = n, \quad \dim \tau = n - 1.$$

Thus τ is a codimension-one face of σ , σ is maximal, and τ is a face of no simplex other than σ .

Step 1: $L = K \setminus \{\sigma, \tau\}$ is again a simplicial complex.

We must show that L is closed under taking faces.

Let $\eta \in L$, and let $\rho \subseteq \eta$ be a face. Since K is a simplicial complex, $\rho \in K$. We must show that $\rho \neq \sigma$ and $\rho \neq \tau$.

First, $\rho \neq \sigma$, since $\rho \subseteq \eta$ and $\eta \neq \sigma$, while σ cannot be a face of a proper sub-simplex.

Next, suppose $\rho = \tau$. Then $\tau \subseteq \eta$. Since $\eta \neq \tau$ and τ is assumed to be a face of no simplex other than σ , it would follow that $\eta = \sigma$, contradicting $\eta \in L$.

Thus $\rho \notin \{\sigma, \tau\}$, so $\rho \in L$. Therefore L is again a simplicial complex.

Step 2: the inclusion $V_L \rightarrow V_K$ defines a simplicial map $i : L \rightarrow K$.

The vertex set of L is contained in the vertex set of K , so there is a natural inclusion

$$V_L \hookrightarrow V_K.$$

If $\eta \in L$, then η is certainly also a simplex of K . Hence the image of every simplex of L is a simplex of K . So the inclusion of vertex sets defines a simplicial map

$$i : L \rightarrow K.$$

(i) Constructing a chain homotopy inverse.

We now show that the induced chain map

$$i_{\#} : C_{\bullet}(L) \rightarrow C_{\bullet}(K)$$

is a chain homotopy equivalence.

Step 3: choose orientations.

Choose orientations of σ and τ so that τ appears in $\partial\sigma$ with coefficient $+1$. Then we can write

$$\partial\sigma = \tau + \beta,$$

where $\beta \in C_{n-1}(L)$ is the sum of the other codimension-one faces of σ , with the appropriate signs.

Indeed, all faces of σ other than τ belong to L , because only σ and τ were removed.

Step 4: define a chain map $r : C_\bullet(K) \rightarrow C_\bullet(L)$.

We define r on oriented simplices and extend linearly.

- If η is a simplex of L and $\eta \neq \tau$, define

$$r(\eta) = \eta.$$

- Define

$$r(\sigma) = 0.$$

- Define

$$r(\tau) = -\beta.$$

Since $\beta \in C_{n-1}(L)$, this is well-defined and lands in $C_\bullet(L)$.

Step 5: verify that r is a chain map.

We must show that

$$r\partial = \partial r.$$

Case 1: $\eta \in L$, $\eta \neq \tau, \sigma$.

Then $r(\eta) = \eta$. Since L is a subcomplex, every face of η also lies in L . Therefore

$$r(\partial\eta) = \partial\eta = \partial(r(\eta)).$$

Case 2: $\eta = \sigma$.

We have

$$r(\sigma) = 0, \quad \partial r(\sigma) = 0.$$

On the other hand,

$$\partial\sigma = \tau + \beta,$$

so

$$r(\partial\sigma) = r(\tau) + r(\beta) = (-\beta) + \beta = 0.$$

Hence

$$r(\partial\sigma) = \partial r(\sigma).$$

Case 3: $\eta = \tau$.

By definition,

$$r(\tau) = -\beta,$$

so

$$\partial r(\tau) = -\partial\beta.$$

Now from

$$\partial\sigma = \tau + \beta$$

and $\partial^2 = 0$, we get

$$0 = \partial^2\sigma = \partial(\tau + \beta) = \partial\tau + \partial\beta.$$

Hence

$$\partial\beta = -\partial\tau,$$

and therefore

$$\partial r(\tau) = -(-\partial\tau) = \partial\tau.$$

Since every proper face of τ lies in L , we have $r(\partial\tau) = \partial\tau$. Thus

$$r(\partial\tau) = \partial r(\tau).$$

So in all cases,

$$r\partial = \partial r.$$

Therefore r is a chain map.

Step 6: $r \circ i_{\#} = \text{id}_{C_{\bullet}(L)}$.

If $c \in C_{\bullet}(L)$, then $i_{\#}(c)$ is simply the same chain viewed inside $C_{\bullet}(K)$, and r fixes every simplex of L . Therefore

$$r \circ i_{\#} = \text{id}_{C_{\bullet}(L)}.$$

Step 7: construct a chain homotopy between $i_{\#}r$ and $\text{id}_{C_{\bullet}(K)}$.

Define

$$h : C_j(K) \rightarrow C_{j+1}(K)$$

on basis simplices by

$$h(\tau) = \sigma,$$

and

$$h(\eta) = 0 \quad \text{for all simplices } \eta \neq \tau.$$

Extend linearly.

We claim that

$$\partial h + h\partial = \text{id}_{C_{\bullet}(K)} - i_{\#}r.$$

We verify this on basis simplices.

Case 1: $\eta \in L$, $\eta \neq \tau$.

Then $h(\eta) = 0$. Also, because τ is not a face of any simplex of L , the boundary $\partial\eta$ contains no τ -term. Hence

$$h(\partial\eta) = 0.$$

So

$$(\partial h + h\partial)(\eta) = 0.$$

On the other hand, $r(\eta) = \eta$, so

$$(\text{id} - i_{\#}r)(\eta) = \eta - \eta = 0.$$

Case 2: $\eta = \sigma$.

Since $h(\sigma) = 0$,

$$\partial h(\sigma) = 0.$$

And because

$$\partial\sigma = \tau + \beta,$$

we get

$$h(\partial\sigma) = h(\tau + \beta) = \sigma.$$

Therefore

$$(\partial h + h\partial)(\sigma) = \sigma.$$

Also $r(\sigma) = 0$, so

$$(\text{id} - i_{\#}r)(\sigma) = \sigma.$$

Case 3: $\eta = \tau$.

We have

$$h(\tau) = \sigma,$$

so

$$\partial h(\tau) = \partial\sigma = \tau + \beta.$$

Since every face of τ lies in L , and h vanishes on simplices other than τ , we have

$$h(\partial\tau) = 0.$$

Thus

$$(\partial h + h\partial)(\tau) = \tau + \beta.$$

On the other hand,

$$r(\tau) = -\beta,$$

so

$$(\text{id} - i_{\#}r)(\tau) = \tau - (-\beta) = \tau + \beta.$$

Hence in all cases,

$$\partial h + h\partial = \text{id}_{C_{\bullet}(K)} - i_{\#}r.$$

So $i_{\#}r$ is chain homotopic to the identity on $C_{\bullet}(K)$.

Together with

$$ri_{\#} = \text{id}_{C_{\bullet}(L)},$$

this shows that $i_{\#}$ and r are chain homotopy inverses.

Step 8: conclude on homology.

Chain homotopy equivalent chain complexes have isomorphic homology. Therefore

$$i_* : H_j(L) \xrightarrow{\cong} H_j(K) \quad \text{for all } j.$$

This proves part (i).

(ii) Proof using the previous question twice.

We now prove the same result using the previous problem, which described the effect of deleting a maximal simplex.

Let

$$K_1 = K \setminus \{\sigma\}.$$

Since σ is not a proper face of any simplex of K , the previous problem applies to the pair

$$K_1 \subset K.$$

Thus there is an exact sequence

$$0 \rightarrow H_n(K_1) \rightarrow H_n(K) \rightarrow \mathbb{Z} \rightarrow H_{n-1}(K_1) \rightarrow H_{n-1}(K) \rightarrow 0,$$

and

$$H_j(K_1) \cong H_j(K) \quad \text{for } j \neq n, n-1.$$

Now observe that in K_1 , the simplex τ is no longer a proper face of any simplex, because its only coface in K was σ , and σ has been removed. Therefore the previous problem applies again, this time to

$$L = K_1 \setminus \{\tau\}.$$

Since $\dim \tau = n-1$, we obtain a second exact sequence

$$0 \rightarrow H_{n-1}(L) \rightarrow H_{n-1}(K_1) \rightarrow \mathbb{Z} \rightarrow H_{n-2}(L) \rightarrow H_{n-2}(K_1) \rightarrow 0,$$

and

$$H_j(L) \cong H_j(K_1) \quad \text{for } j \neq n-1, n-2.$$

Now the key point is that these two changes cancel each other, because τ is a free face of σ . Algebraically, removing σ creates a possible change controlled by the class of $\partial\sigma$, and removing τ immediately removes exactly that remaining contribution. Geometrically, the pair (σ, τ) is an elementary collapse pair, so deleting both should not change homology.

Thus the composite inclusion

$$L \hookrightarrow K_1 \hookrightarrow K$$

induces an isomorphism

$$H_j(L) \xrightarrow{\cong} H_j(K) \quad \text{for all } j.$$

This proves part (ii).

15.4 Conclusion

We have shown the following.

- If σ is a maximal simplex and $\tau < \sigma$ is a codimension-one face which is not a face of any other simplex, then

$$L = K \setminus \{\sigma, \tau\}$$

is again a simplicial complex.

- The inclusion of vertex sets defines a simplicial map

$$i : L \rightarrow K.$$

- The induced chain map

$$i_{\#} : C_{\bullet}(L) \rightarrow C_{\bullet}(K)$$

is a chain homotopy equivalence.

- Consequently,

$$i_* : H_j(L) \xrightarrow{\cong} H_j(K) \quad \text{for all } j.$$

- The same conclusion may also be derived by applying the previous deletion result twice: first delete σ , then delete τ .

Thus deleting a free face together with the unique simplex containing it preserves homology in all degrees. This is the algebraic-topological content of an elementary collapse.

15.5 Examples of elementary collapses

We now give several concrete examples illustrating the situation of the previous problem.

Recall the setup: K is a simplicial complex, $\sigma \in K$ is a simplex which is not a proper face of any simplex, and $\tau < \sigma$ is a codimension-one face which is not a face of any other simplex. Then

$$L = K \setminus \{\sigma, \tau\}$$

is obtained from K by an *elementary collapse*, and the inclusion

$$i : L \hookrightarrow K$$

induces isomorphisms

$$i_* : H_j(L) \xrightarrow{\cong} H_j(K) \quad \text{for all } j.$$

Example 1: collapsing a single edge to a point.

Let K consist of one edge $[v_0v_1]$ together with its two vertices $[v_0]$ and $[v_1]$. Take

$$\sigma = [v_0v_1], \quad \tau = [v_1].$$

Then σ is maximal, τ is a codimension-one face of σ , and τ is not a face of any other simplex. Thus

$$L = K \setminus \{[v_0v_1], [v_1]\}$$

is the complex consisting of the single vertex $[v_0]$.

Geometrically, one has

$$v_0 - v_1 \rightsquigarrow v_0.$$

Both K and L are contractible, so

$$H_j(K) \cong H_j(L) \quad \text{for all } j.$$

Example 2: collapsing a filled triangle along a free edge.

Let K be the full 2-simplex $[abc]$, together with all of its faces. Take

$$\sigma = [abc], \quad \tau = [ab].$$

Then $[ab]$ is a codimension-one face of $[abc]$, and since $[abc]$ is the only 2-simplex in K , the edge $[ab]$ is not a face of any other simplex of dimension 2. Therefore

$$L = K \setminus \{[abc], [ab]\}$$

is the complex consisting of the two edges $[ac]$, $[bc]$, their vertices, and no 2-simplex.

Geometrically, the filled triangle collapses to a V -shaped graph:

$$\text{filled triangle} \rightsquigarrow [ac] \cup [bc].$$

Both spaces are contractible, so again

$$H_j(K) \cong H_j(L) \quad \text{for all } j.$$

This example is particularly useful because one can also see the chain homotopy explicitly. Choosing orientations so that

$$\partial[abc] = [ab] - [ac] + [bc],$$

we have

$$\partial\sigma = \tau + \beta \quad \text{with} \quad \beta = -[ac] + [bc].$$

Hence in the proof of the theorem one may define

$$r([abc]) = 0, \quad r([ab]) = -\beta = [ac] - [bc],$$

and let r fix all remaining simplices.

Example 3: a square cut into two triangles.

Let K be a square triangulated by one diagonal. Thus K has vertices a, b, c, d , edges

$$[ab], [bc], [cd], [da], [ac],$$

and 2-simplices

$$[abc], \quad [acd].$$

Take

$$\sigma = [abc], \quad \tau = [ab].$$

Then $[ab]$ is a face of $[abc]$, and it is not a face of $[acd]$. Hence $[ab]$ is a free face of $[abc]$, and

$$L = K \setminus \{[abc], [ab]\}$$

is the remaining triangle $[acd]$ together with its faces.

So the square triangulated by a diagonal collapses to a single triangle. Since both are contractible, their homology groups are the same.

Example 4: collapsing a tetrahedron along one triangular face.

Let K be the full 3-simplex $[abcd]$, together with all of its faces. Take

$$\sigma = [abcd], \quad \tau = [abc].$$

Then τ is a codimension-one face of σ , and since $[abcd]$ is the only 3-simplex, τ is not a face of any other simplex. Thus

$$L = K \setminus \{[abcd], [abc]\}$$

is the union of the remaining three triangular faces and all of their subfaces.

Geometrically, one removes the solid tetrahedron together with one of its boundary faces, and what remains is a contractible complex. Thus

$$H_j(K) \cong H_j(L) \quad \text{for all } j.$$

This is the 3-dimensional analogue of the collapse of a filled triangle along a free edge.

Example 5: pruning a leaf edge from a graph.

Let K be a graph consisting of a circle with one extra dangling edge attached at a point. Let the dangling edge be $\sigma = [uv]$, where v is the leaf vertex. Take

$$\tau = [v].$$

Then τ is a codimension-one face of the 1-simplex σ , and since v is a leaf, it is not a face of any other edge. Hence

$$L = K \setminus \{[uv], [v]\}$$

is the circle.

Thus the graph with a hanging branch collapses to the circle. In particular,

$$H_0(K) \cong H_0(L) \cong \mathbb{Z}, \quad H_1(K) \cong H_1(L) \cong \mathbb{Z},$$

and $H_j = 0$ for $j \geq 2$.

This is a useful example because it shows that one may repeatedly prune leaf edges from a graph without changing homology.

Example 6: repeated collapses of a triangle to a point.

Start with the full 2-simplex $[abc]$. First collapse the pair

$$([abc], [ab]).$$

This leaves the V-shaped complex $[ac] \cup [bc]$. Next collapse

$$([ac], [a]),$$

leaving the single edge $[bc]$. Finally collapse

$$([bc], [b]),$$

leaving only the vertex $[c]$.

So one obtains a sequence of elementary collapses

$$[abc] \rightsquigarrow [ac] \cup [bc] \rightsquigarrow [bc] \rightsquigarrow [c].$$

At each step the homology remains unchanged, so the original filled triangle has the same homology as a point, namely

$$H_0 \cong \mathbb{Z}, \quad H_j = 0 \text{ for } j \geq 1.$$

This example shows how repeated elementary collapses provide a concrete combinatorial method for proving that a complex is contractible.

Example 7: a non-example — a triangulated circle.

Let K be a circle triangulated by three edges:

$$[ab], \quad [bc], \quad [ca].$$

Take $\sigma = [ab]$. Its codimension-one faces are $[a]$ and $[b]$. But each of these vertices is also a face of another edge in the circle. Therefore neither $[a]$ nor $[b]$ is a free face.

So Problem 5 does *not* apply. Indeed, the circle cannot be collapsed to a point by elementary collapses. This is consistent with homology:

$$H_1(K) \cong \mathbb{Z},$$

so K is not contractible.

This non-example is important because it shows that not every simplex can be removed without changing topology; the existence of a free face is essential.

Example 8: peeling boundary triangles from a triangulated disk.

Let K be a triangulated 2-disk. If a boundary triangle has an edge lying on the outer boundary of the disk and not shared by any other triangle, then that boundary edge is a free face. Hence one may remove the boundary triangle together with that edge.

By repeating this process, one can often collapse the whole triangulated disk to a single vertex. Therefore a triangulated disk has the same homology as a point:

$$H_0 \cong \mathbb{Z}, \quad H_j = 0 \text{ for } j \geq 1.$$

This is a fundamental geometric source of elementary collapses.

15.6 Summary of the examples

These examples show the basic geometric meaning of Problem 5:

- a simplex together with a free codimension-one face may be deleted without changing homology;
- in graphs, this corresponds to pruning leaf edges;
- in triangulated disks or balls, this corresponds to peeling off boundary simplices;

- repeated elementary collapses often reduce a contractible complex to a point;
- the hypothesis that the chosen face be free is essential.

Thus Problem 5 is the algebraic-topological expression of the fact that elementary collapses preserve homology.

15.7 Examples of simplicial approximation

We now give several concrete examples illustrating the simplicial approximation theorem.

Recall the theorem in its basic form: if K and L are simplicial complexes and

$$f : |K| \rightarrow |L|$$

is continuous, then after sufficiently many barycentric subdivisions of K , there exists a simplicial map

$$g : \text{sd}^r K \rightarrow L$$

such that

$$|g| \simeq f.$$

Thus every continuous map between polyhedra is, up to homotopy, represented by a simplicial map from a sufficiently fine subdivision of the domain.

Example 1: the identity map on an interval.

Let $K = L$ be the simplicial complex consisting of one edge $[v_0v_1]$ and its two vertices. Then

$$|K| = |L| = [0, 1].$$

The identity map

$$f : [0, 1] \rightarrow [0, 1], \quad f(x) = x$$

is already simplicial: it sends each vertex to itself and the unique edge to itself. So in this case no subdivision is needed, and one may take

$$g = f.$$

This is the simplest possible example: a continuous map which is already simplicial.

Example 2: a constant map on a triangle.

Let K be a triangle and L another simplicial complex. Suppose

$$f : |K| \rightarrow |L|$$

is constant, with image equal to some vertex $w \in L$. Then f is homotopic to the simplicial map $g : K \rightarrow L$ defined on vertices by

$$g(v) = w \quad \text{for every vertex } v \in K.$$

Since the image of every simplex is the single vertex w , this is a simplicial map.

Thus constant maps are immediate examples of simplicial approximations.

Example 3: the identity map on a triangulated circle.

Triangulate S^1 by three vertices v_0, v_1, v_2 and three edges

$$[v_0v_1], \quad [v_1v_2], \quad [v_2v_0].$$

Likewise triangulate another copy of S^1 by vertices w_0, w_1, w_2 and the analogous edges.

The identity map of the circle can then be realized simplicially by defining

$$g(v_i) = w_i \quad (i = 0, 1, 2).$$

This sends edges to edges and induces a simplicial map

$$g : S^1 \rightarrow S^1$$

of degree 1.

So the identity map on a circle is simplicial once compatible triangulations are chosen.

Example 4: a degree 2 map $S^1 \rightarrow S^1$.

Let

$$f : S^1 \rightarrow S^1$$

be the map of degree 2, which wraps the circle twice around itself.

With a coarse triangulation of the domain, such a map is usually not simplicial: one edge of the domain may traverse too much of the target circle. However, after subdividing the domain circle sufficiently many times, one can define a simplicial approximation.

For instance, suppose the domain has vertices

$$v_0, v_1, v_2, v_3, v_4, v_5$$

cyclically, and the target has vertices

$$w_0, w_1, w_2$$

cyclically. Define

$$g(v_0) = w_0, \quad g(v_1) = w_1, \quad g(v_2) = w_2, \quad g(v_3) = w_0, \quad g(v_4) = w_1, \quad g(v_5) = w_2.$$

Then the induced simplicial map travels twice around the target circle, so it has degree 2.

This example shows clearly why subdivision of the domain is often necessary.

Example 5: a nonlinear map of the interval requiring subdivision.

Let $K = L = [0, 1]$ with the usual triangulation by one edge. Consider the continuous map

$$f : [0, 1] \rightarrow [0, 1], \quad f(x) = 4x(1 - x).$$

This map is not simplicial with respect to the given triangulations. Indeed, the endpoints satisfy

$$f(0) = 0, \quad f(1) = 0,$$

while the midpoint satisfies

$$f\left(\frac{1}{2}\right) = 1.$$

So the image of the single edge bends upward and then back down; it is not the realization of a simplicial map on the original triangulation.

However, after subdividing the interval into many small edges, the map can be approximated simplicially. This is one of the standard motivating examples: the map is continuous, but not simplicial on a coarse triangulation, and subdivision resolves the problem.

Example 6: a map $S^1 \rightarrow S^2$.

Let

$$f : S^1 \rightarrow S^2$$

be any continuous map. Choose triangulations

$$|K| \cong S^1, \quad |L| \cong S^2.$$

By simplicial approximation, after subdividing K sufficiently many times, f is homotopic to a simplicial map

$$g : \text{sd}^r K \rightarrow L.$$

Since the domain is 1-dimensional, the image of g lies in the 1-skeleton $L^{(1)}$, which is only a graph inside the sphere S^2 . In particular, the image cannot be all of S^2 , so there is some point

$$y \in S^2$$

not in the image of g . Hence

$$g : S^1 \rightarrow S^2 \setminus \{y\}.$$

But

$$S^2 \setminus \{y\} \cong \mathbb{R}^2,$$

and $\mathbb{R}^2$ is contractible. Therefore g , and hence f , is null-homotopic.

This is a central example of how simplicial approximation is used to prove

$$\pi_1(S^2) = 0.$$

Example 7: maps $S^m \rightarrow S^n$ with $m < n$.

More generally, let

$$f : S^m \rightarrow S^n$$

with $m < n$. After choosing triangulations and applying simplicial approximation, one obtains a simplicial map

$$g : \text{sd}^r K \rightarrow L$$

with

$$|K| \cong S^m, \quad |L| \cong S^n,$$

and $g \simeq f$.

Because the domain has dimension m , the image of g lies in the m -skeleton $L^{(m)}$. Since $m < n$, this is a proper subcomplex of L , so the image misses a point of S^n . Therefore g , and hence f , is null-homotopic.

This is the standard simplicial approximation proof that every map

$$S^m \rightarrow S^n \quad (m < n)$$

is homotopic to a constant map.

Example 8: pushing loops into the 1-skeleton.

Let K be any simplicial complex, and let

$$\alpha : S^1 \rightarrow |K|$$

be a loop. Triangulate S^1 as a 1-dimensional simplicial complex. By simplicial approximation, after subdivision, α is homotopic to a simplicial map

$$g : \text{sd}^r(S^1) \rightarrow K.$$

Since the domain is 1-dimensional, the image of g lies in the 1-skeleton $K^{(1)}$. Therefore every loop in $|K|$ is homotopic to a loop in $|K^{(1)}|$.

This is a fundamental application of the theorem in the study of the fundamental group.

Example 9: pushing homotopies into the 2-skeleton.

Suppose two loops in $|K|$ are homotopic, so there is a homotopy

$$H : S^1 \times I \rightarrow |K|.$$

Triangulate the cylinder $S^1 \times I$ as a 2-dimensional simplicial complex, and after sufficient subdivision apply simplicial approximation to obtain a simplicial map

$$H' : \text{sd}^r(S^1 \times I) \rightarrow K$$

homotopic to H .

Since the domain is 2-dimensional, the image of H' lies in the 2-skeleton $K^{(2)}$. Therefore homotopies between loops may be taken inside $|K^{(2)}|$.

This is the essential simplicial approximation argument behind the fact that

$$\pi_1(|K^{(2)}|) \cong \pi_1(|K|).$$

Example 10: a map from a triangle to a circle.

Let K be a triangle and let L be a triangulated circle. Consider a continuous map

$$f : |K| \rightarrow |L|.$$

Since $|K|$ is contractible, every such map is null-homotopic. By simplicial approximation, after subdividing K , the map is homotopic to a simplicial map

$$g : \text{sd}^r K \rightarrow L.$$

Because the target L is 1-dimensional, every 2-simplex of $\text{sd}^r K$ must map into an edge or a vertex of L . Thus one sees geometrically that a 2-dimensional region may be folded combinatorially onto a graph.

This is a useful illustration that simplicial maps do not preserve dimension.

Example 11: simplicial approximation in no-retraction arguments.

Suppose one tries to define a retraction

$$r : D^2 \rightarrow S^1$$

from the disk onto its boundary. If such a map existed, then after triangulating the disk and the circle and subdividing the domain sufficiently many times, simplicial approximation would produce a simplicial map

$$g : \text{sd}^r(D^2) \rightarrow S^1$$

homotopic to r , with boundary behavior matching the identity on S^1 .

This combinatorial reformulation is often the starting point for impossibility proofs, such as no-retraction theorems and Brouwer-type arguments. Thus simplicial approximation is not only constructive; it is also a powerful tool for proving that certain maps cannot exist.

Example 12: the geometric mechanism behind the theorem.

The proof of the simplicial approximation theorem is guided by the *star condition*. After sufficient subdivision of the domain, one can ensure that for each vertex v of $\text{sd}^r K$, there exists a vertex w of L such that

$$f(\text{St}(v)) \subseteq \text{St}(w).$$

Then one defines the simplicial approximation by

$$g(v) = w.$$

Thus one should picture the theorem in the following way:

- subdivide the domain until its simplices are very small;
- on the star of each small vertex, the image under f fits inside the star of some target vertex;
- choose that target vertex as the image of the domain vertex;
- the resulting vertex assignment defines a simplicial map.

This is the geometric reason simplicial approximation works.

15.8 Summary

These examples illustrate the main uses of simplicial approximation:

- replacing continuous maps by combinatorial ones after subdivision;
- constructing simplicial representatives of maps of circles and spheres;
- proving null-homotopy results such as

$$S^m \rightarrow S^n \quad (m < n);$$

- pushing loops into the 1-skeleton and homotopies into the 2-skeleton;
- reformulating existence and nonexistence questions in combinatorial terms.

Thus simplicial approximation is one of the key bridges between topology and the combinatorics of simplicial complexes.